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Dynamic multi-swarm whale optimization algorithm based on elite tuning for high-dimensional feature selection classification problems

Fahui Miao [a], Yong Wu [b], Guanjie Yan [c,*], Xiaomeng Si [d]

[a] College of Information Engineering, Shanghai Maritime University, Shanghai 201306, China
[b] University of Macau Zhuhai UM Science and Technology Research Institute, Zhuhai 519000, China
[c] The Department of Basic Education, Shanghai Urban Construction Vocational College, Shanghai 201415, China
[d] School of Computer and Information Engineering, Institute for Artificial Intelligence, Shanghai Polytechnic University, Shanghai 201209, China

HIGHLIGHTS

- Proposed a dynamic multi-swarm construction mechanism based on centroid distance.
- An elite tuning strategy to address invalid probability flips is proposed.
- Dynamic multi-swarm whale optimization algorithm based on elite tuning is proposed.
- The proposed algorithm is used for high-dimensional feature selection problems.
- A superior performance of the proposed algorithm in feature selection is proved.

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ABSTRACT

Feature selection algorithms are crucial technologies for reducing the dimensionality of high-dimensional data. However, the exponential expansion of the decision space due to high-dimensional data makes most feature selection algorithms highly susceptible to local optima. To increase the quality of feature selection, we propose a dynamic multi-swarm whale optimization algorithm based on elite tuning (EMSWOA). First, we propose a dynamic multiple swarm construction mechanism based on the centroid distance metric, which enhances the exploitation and exploration abilities of the algorithm by improving the connectivity between sub-population topologies and preventing the algorithm from falling into precocious convergence. Secondly, to address the issue of invalid probability flips caused by random thresholds during feature space transformation, we design a novel elite tuning mechanism. This mechanism uses the high-quality information in the elite solution to assess and correct the probability of feature flipping; it effectively improves the algorithm's recognition of important features and reduces the interference of invalid probability flips in the optimization search. Finally, we validated the EMSWOA against six comparison algorithms on 19 datasets. The experimental results demonstrate that, compared with the most effective benchmark algorithms, the EMSWOA yields an average accuracy increase of 2.91 % and an average reduction in the number of features by a factor of 3.58.

1. Introduction

With the advent of the Big Data era, an increasing amount of high-dimensional data are appearing in everyday life. However, the high-dimensional data collected contain considerable redundant or irrelevant information, which not only increases the computational overhead of the model, but also causes degradation in model performance due to problems such as over-fitting [1]. Feature Selection (FS) is an effective dimensionality reduction method that reduces the complexity of features, decreases computational overhead, and enhances accuracy by selecting the optimal subset from high-dimensional feature data, thereby eliminating redundant and irrelevant features [2]. As a key technique for data preprocessing, feature selection algorithms are widely used in numerous practical problems such as neural data analysis

* Correspondence to: The Department of Basic Education, Shanghai Urban Construction Vocational College, No. 2080, Nanting Highway, Fengxian District, Shanghai 200438, China.
E-mail addresses: miaofahui@126.com (F. Miao), zumri.ywu@um.edu.mo (Y. Wu), yanguanjiesucc@163.com (G. Yan), sixm@sspu.edu.cn (X. Si).

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[3].

Feature selection algorithms include filter-based approaches, wrapper-based approaches and embedded approaches [4]. Filter-based algorithms usually filter a subset by the relevance criterion of the features, and therefore have a low computational overhead [5]. As the evaluation criteria of filter-based algorithms are not related to the learning process of the algorithm, the accuracy of the selected subset performs poorly. Wrapper-based feature selection algorithms, however, generally treat the feature selection process as a search problem. They identify candidate feature subsets using methods such as heuristic searches and directly employ classifier performance metrics to evaluate the merits of these subsets. The optimal feature subset is obtained by integrating search algorithms with classifiers trained during the process. This type of method, taking into account the correlations among features, imposes higher demands on the global optimization capabilities of search algorithms [6]. Additionally, embedded feature selection methods integrate the feature selection process into the model training, leveraging the model's inherent mechanisms to automatically perform feature selection through model parameters or structures. However, the effectiveness of feature selection is directly dependent on the choice of model, potentially constrained by the model's own performance [7].

Although feature selection algorithms are continuously evolving, most methods still face numerous challenges when performing classification tasks on high-dimensional datasets [8]. Consequently, the task of high-dimensional feature selection remains a focal point of research. The FS problem is essentially a combinatorial optimization issue [9]. The goal of the FS is to effectively remove redundant and irrelevant features to obtain a higher classification accuracy with a minimum number of features. Metaheuristic algorithms have shown outstanding capabilities for addressing combinatorial optimization issues and have been successful in recent years for combinatorial optimization problems such as data clustering [10,11]. Benefiting from the global optimization capabilities of the meta-heuristic algorithm, it has also gained significant attention. In contrast to approaches such as exhaustive searching [12], metaheuristics do not rely on a priori knowledge of the domain or specific assumptions about the search space, making them more suitable for feature selection tasks [13].

The whale optimization algorithm (WOA) [14] has received much attention as a representative algorithm in the field of metaheuristic algorithms in recent years. The WOA is suitable for solving complex feature selection problems in the search space because of its strong global optimization capability [15] and flexible search direction control mechanism. Specifically, the vector representation of individual whales in the population is consistent with the size of the feature subset of the selected features. In the WOA, the feeding process of whales is the random search process of constructing a subset of features in feature selection. Individual whales with better fitness values during WOA predation were considered more important for the selected traits. A subset of the more optimal features was obtained through the flexible foraging process of the whales. In addition, more importantly, due to the shrinking encircling mechanism and spiral updating position of the WOA algorithm, the WOA algorithm can more flexibly control search direction in complex space; thus, it can be better realized in complex space spatial exploitation and exploration. Therefore, the WOA algorithm may be more suitable for solving the feature selection problem. WOA algorithms have been tested on high-dimensional and large-scale optimization problems. For example, Zhang et al. [16] proposed a hybrid WOA based on aggregation strategies for high-dimensional problems.

However, the above WOA-based feature algorithms, which are dominated by single populations [17], are still unable to overcome the problem of insufficient exploration capability of high-dimensional complex solution spaces [18]. Due to the computational time constraints inherent in the single-swarm Whale Optimization Algorithm (WOA) for feature selection, it is challenging to conduct unlimited searches and iterations over feature subsets. Consequently, the single-population WOA typically explores only a small part of the decision space, which significantly restricts the acquisition of the optimal subset and adversely impacts the quality of the final feature selection. Therefore, it is crucial to improve the search capability of WOA algorithms in high-dimensional feature spaces. Multi-swarm optimization algorithms can divide the entire population into multiple sub-swarms, which learn and update in their own search spaces. This helps to improve the diversity of the population, increase the exploration ability, and can effectively avoid premature convergence. For example, Yang et al. [19] proposed an evolutionary state-driven multi-population cooperative particle swarm optimization algorithm for complex optimization problems.

In addition, none of the above feature selection algorithms have been improved for the invalid probability flipping problem. This is because existing metaheuristic feature selection algorithms employ transformation functions that are usually of two types, S-type or V-type [3]. By applying the transformation function to the update step size in each iteration cycle, the corresponding probability value for each dimension is obtained. This probability is also used as a measure of feature importance, and the feature is flipped or selected based on a comparison of the probability value with a threshold. Therefore, setting the threshold is crucial for determining which features are selected. There are two existing methods for setting the threshold: a fixed threshold based on a constant value and a dynamic threshold based on random values. Using a constant value as the threshold is not only make it difficult to determine the optimal setting but also results in a lack generalizability for different datasets. For example, Yousri et al. [20] used a fixed threshold in their marine predator feature selection algorithm. Using a dynamic threshold for feature importance measurement can increase the algorithm's perturbation, but there is also a problem of invalid probability flipping due to the randomness of the dynamic threshold; for example, in Abualigah et al.'s [21] used a random threshold for feature measurement. Invalid probability flipping can reduce the selection probability of high-quality features, thereby affecting the search speed and quality of the best subset.

To overcome the insufficient detection ability of the WOA algorithm in high-dimensional complex solution space and the problem of invalid probability flipping when features are mapped discretely, we propose a dynamic multi-swarm WOA based on elite tuning (EMSWOA). Building on the foundation of the WOA algorithm, we introduce an elite correction strategy to alleviate the problem of invalid probability flipping and devise a dynamic multi-swarm construction mechanism based on centroid distance to enhance the global optimization ability of the EMSWOA and improve the quality of feature subsets. Our innovations are as follows:

- A dynamic multi-swarm strategy based on centroid distance was proposed, which uses centroid measurements to directionally guide the dynamic formation of subpopulations, enhancing their exploitation capabilities.
- A dynamic population recombination mechanism with a linearly decreasing number of subpopulations was developed to increase the algorithm's independence in early search stages within the feature space and enhance information exchange and global exploration capabilities in later iterations.
- A local sparsity metric mechanism was introduced, utilizing a comprehensive assessment of individual congestion and dispersion in the feature space to maintain the diversity of candidate solutions, thereby increasing the probability of identifying the optimal feature subset.
- An elite tuning strategy is proposed to use two generations of elite individuals to correct the probabilistic invalid flip problem caused by random thresholds and to avoid the performance drain and disturbance caused by invalid probabilistic flips.

- The EMSWOA is proposed and compared with other prominent feature selection algorithms on high-dimensional datasets, demonstrating the superiority of the EMSWOA algorithm.

The organization is as follows. Section 2 reviews the relevant literature, including the principles of feature selection and the WOA. The EMSWOA algorithm is described in detail in Section 3. Section 4 presents the experimental results and discussion, which verify and analyse the performance of the proposed algorithm. Section 5 summarizes the paper and provides a vision of future research trends.

2. Related work

2.1. Feature selection

The essence of the FS problem lies in a problem of compositional optimization [3]. The general form of the definition of a single-objective optimization problem is as follows:

$$\begin{aligned} &\min\ f(x) \\ &s.t.\quad X=(x_1,x_2,\cdots,x_N) \end{aligned} \tag{1}$$

where $f(x)$ denotes the objective function to be optimized and x_N denotes the Nth dimension of the candidate solution.

The purpose of feature selection is to select the best subset of features from a dataset containing N features to improve the accuracy of the algorithm. Therefore, the objective function of the model for the feature selection problem is defined such that it is desirable to minimize both the classification error and minimize the number of features in the feature subset. We can represent the feature selection problem via the following formula [8]:

$$\begin{aligned} &\min\ f(x)=\alpha * ErrorRate+\beta * \frac{\#Selected}{\#All} \\ &s.t.\quad X=(x_1,x_2,\cdots,x_N) \\ &\qquad x_i\in\{0,1\},\forall i\in\{1,2,3,\cdots,N\} \end{aligned} \tag{2}$$

where *ErrorRate* denotes the classification error, $\#Selected$ denotes the number of features in the feature subset, and #All denotes the number of all features contained in the original feature. The sum of α and β is 1, which is the coupling factor for achieving both objectives simultaneously. X represents the set containing all the features, Nrepresents feature dimensions, and 0 and 1 are used to indicate whether the i-th feature is selected. $f(x)$ represents the feature evaluation function, which is typically represented by the classification error and is used to measure the quality of the selected subset. By using $f(x)$ to guide the feature selection, we obtain the best subset with the minimum error.

2.2. Whale optimization algorithm (WOA)

The WOA is a novel swarm intelligence algorithm proposed by Mirjalili et al. and inspired by the unique hunting process of whales [14]. During the hunting process, whales surround their prey by forming a cylindrical or tubular bubble net with spiralling bubbles, continuously forcing the prey towards the centre and making final capture easier. This process is also referred to as bubble-net hunting. The WOA is based on this hunting process and is abstracted into three predation stages, namely, encircling prey, bubble-net attacking and random searching. The formula for the WOA is as follows:

2.2.1. Encircling prey

During the encircling prey phase of whale feeding, the whales' main objective is to observe the location of their prey and encircle it for predation. Information is passed between whales through the population, the position of the nearest whale to the prey is set as the global best solution $X^*$of the current whale, and the search area is narrowed by approximating the current best solution and encircling the prey. The encircling and feeding phase is represented as follows [14]:

$$D=|C\cdot X^*(t)-X(t)| \tag{3}$$

$$C=2\cdot r \tag{4}$$

where Ddenotes the swim step in approximation to the optimal solution and t indicates the current number of iterations. C indicates the oscillation factor, r denotes a stochastic number in [0,1], and $X(t)$denotes the location of the current individual whale in the t-th generation.

A whale in a swarm is location-modified by trying to approximate the optimal individual, which is computed via the following model [14]:

$$X(t+1)=X^*(t)-A\cdot D \tag{5}$$

$$A=2a\cdot r-a \tag{6}$$

where A denotes the convergence factor, a decreases linearly from 2 to 0, and r denotes a stochastic number in [0,1].

2.2.2. Bubble-net attack method (exploitation phase)

During this feeding phase, whales attack their prey in spiral bubbles through both a shrinking encircling mechanism (SEM) and a spiral updating position (SUP) [14]. SEM simulates the gradual reduction in the encircling circle as the whale feeds. The formula for calculating the SEM is the same as Eq. (6), where the range of whale encircling contraction is influenced by the convergence factor A. The size of A is proportional to the search range of the individual whale and is proportional to a. The value of a decreases linearly from 2 to 0, so the range of values of A is [-a,a].

The SUP simulates the spiral swimming of a whale as it feeds, and the formula for the SUP is shown below [14]:

$$X(t+1)=D'\cdot e^{bl}\cdot\cos(2\pi l)+X^*(t) \tag{7}$$

$$D'=|X^*(t)-X(t)| \tag{8}$$

where Ddenotes the swim step of the current individual $X(t)$ as it approaches the optimal solution $X^*(t)$, b is a constant used to control the shape of the spiral, and l denotes a stochastic value between [-1,1].

During the bubble attack phase, both the SEM and the SUP will be present, and assuming a 50 % probability of performing one of these mechanisms, the position updating equation for this phase is shown below [14]:

$$X\left(t+1\right)=\begin{cases} X^*(t)-A\cdot D & if\ \ p<0.5 \\ D'\cdot e^{bl}\cdot\cos(2\pi l)+X^*(t) & if\ \ p\geq 0.5 \end{cases} \tag{9}$$

where p is a stochastic number between [0,1], indicating the probability of choosing a particular behaviour.

2.2.3. Search for prey (exploration phase)

The whale feeding process also includes a random search phase, which simulates the whale's search for prey in a larger space. The individual is not learning from the current best individual, but rather from a stochastic one. The extent of the whale's search is influenced by the convergence factor A. The search range of the solution space is controlled by A. When $|A|>1$, the WOA performs a larger global search, thus enhancing the global search power. The computational model for the random search phase is shown below [14]:

$$X(t+1)=X_{rand}(t)-A\cdot D \tag{10}$$

$$D=|C\cdot X_{rand}(t)-X(t)| \tag{11}$$

where X_{rand} denotes a randomly chosen agent, C is the oscillation factor, and D denotes the distance between the selected random whale $X_{rand}(t)$ and the current whale$X(t)$.

2.3. Literature review of feature selection methods

Based on the feature selection process, feature selection algorithms can be categorized into evolutionary computation (EC) class methods and non-evolutionary computation (NON-EC) class methods. This section provides a literature review of these two categories of feature selection algorithms.

2.3.1. EC-based feature selection

Due to the strong global optimization capabilities of EC algorithms, EC-based feature selection methods have long been among the commonly used approaches for feature dimensionality reduction [22]. However, given the exponential growth in feature space complexity due to high-dimensional data, exploring diversity in EC feature selection methods is highly important. One line of research enhances the global optimization capability of EC algorithms to improve their accuracy in feature selection [23–26]. For instance, Hu et al. [23] introduced an enhanced black widow optimization algorithm for feature selection by designing new spider mate and mutation strategies. Mohamed et al. [24] enhanced the capabilities of feature selection by integrating a chaos optimization strategy with multiple optimization mechanisms, proposing a binary-enhanced Spider Wasp Optimizer. Nonetheless, due to the complexity of high-dimensional feature spaces, methods that merely improve the optimization capabilities of EC algorithms still risk converging to local optima. Some scholars perform feature selection on high-dimensional data by combining filter and EC algorithms. For example, Chen et al. [5] et al. used ReliefF algorithm for initial filtering of high dimensional features followed by particle swarm optimization algorithm for dimensionality reduction of high dimensional features through multi-task information sharing. Askr et al. [27] improved the golden jackal optimization algorithm by combining Copula entropy and four enhancement strategies, achieving dimensionality reduction of features. Pan et al. [28] utilized the ReliefF algorithm and Copula entropy simultaneously to enhance the quality of the initial population, and combined this with an improved Gray Wolf Optimization algorithm to select features from high-dimensional data. However, this hybrid approach to multi-type feature selection increases the complexity of the algorithm and the initial use of a filter-based feature selection method may result in the loss of diversity within the feature subsets. Some scholars have also recognized the potential of multi-swarm structures in high-dimensional feature selection tasks. For instance, Braik et al. [29] proposed an adaptive dynamic elite opposition-based Ali Baba and the forty thieves algorithm, which divides the entire population into two sub-populations to maintain diversity within the population. However, a smaller number of sub-populations might not effectively cover the extensive high-dimensional feature space. To further enhance the extensive exploration of high-dimensional feature spaces, Li et al. [30] introduced a decomposition-based multi-population evolutionary algorithm. This method involves a multi-population generation strategy based on feature weights, which improves the global exploration capability of the algorithm. Nevertheless, the fixed number of sub-populations, being a hyperparameter that does not adapt dynamically during the optimization process, limits the further enhancement of feature selection capabilities.

The WOA as one of the representative algorithm in the EC domain, has been increasingly utilized in the feature selection arena. Recent literature on WOA-based feature selection algorithms is catalogued in Table 1. These algorithms are bifurcated based on the scale of the feature selection datasets into categories targeting conventional and high-dimensional datasets. For conventional-sized feature selection tasks, one line of research focuses on enhancing the feature search capabilities of the WOA [17,31–34], such as the multi-strategy multi-objective WOA proposed by Yang et al., which boosts the feature selection performance of the algorithm [32]. Another line of research combines WOA-based wrappers and filters to accomplish feature selection, such as Yab et al., who enhanced feature selection capabilities by integrating modified WOA with filters [35]. In the context of high-dimensional feature selection challenges, one approach involves augmenting feature recognition capabilities in vast spaces by modifying or amalgamating other algorithms to forge new search strategies. For example, Too et al. restructured the WOA with a spatial boundary strategy, facilitating the identification of latent features in expansive feature spaces [36]. As research progresses, the potency of multi-population structures in complex solution spaces has been recognized, leading to the proposal of multi-population WOA algorithms for intricate feature selection tasks [35–38]. Sun et al., for example, introduced a two-stage adaptive fuzzy neighborhood feature selection algorithm based on WOA, utilizing an adaptive neighborhood search radius to segregate elite and ordinary populations, thereby effectively executing the feature selection process [38]. However, the method has a small number of subpopulations and a fixed number of subpopulations, all of which limit the ability to mine for global information. Current studies utilizing multi-population WOA algorithms for feature selection are scant and warrant further investigation.

Moreover, in recent years, EC feature selection algorithms integrated with attention mechanisms have been proposed. For example, Xue et al. [40] developed a large-scale feature selection ranking algorithm based on an external attention mechanism, which uses an attention module trained during the process to rank features by importance and accomplish feature selection through EC methods. This approach performs well in large-scale feature selection tasks, but it is sensitive to hyperparameters [40].

2.3.2. NON-EC-based feature selection

In recent years, scholars have researched feature selection algorithms for NON-EC, such as ReliefF [41] and mutual information [42]. Zhou et al. [43] proposed a hybrid feature selection method that uses the absolute values of correlation coefficients between features as weights for redundancy in mutual information representation, achieving feature selection. Dai et al. [44] introduced a fuzzy combined entropy feature selection algorithm that simultaneously considers global and local feature relevancies, leveraging both global and local information.

Table 1
Analysis of feature selection algorithms based on WOA in the past four years.

Approaches	Literature	High Dimension	Multi -swarm	Dynamic subpopulation number
Quantum based WOA for wrapper feature selection	Agrawal et al. [31]	X	X	X
Spatial bound whale optimization algorithm	Too et al. [36]	✓	X	X
Hybrid filter-wrapper feature selection using the WOA	Got et al. [39]	X	X	X
An improved WOA (IWOA) and improved KNN for feature selection	Liu et al. [33]	X	X	X
A Modified Whale Optimization Algorithm as Filter-Based	Yab et al. [35]	✓	X	X
Two-stage-fuzzy-neighborhood feature selection with binary whale optimization algorithm	Sun et al. [38]	✓	✓	X
A novel feature selection method based on binary hybrid WOA	Uzer et al. [17]	X	X	X
Improved Whale Optimization with Local-Search	Alzaqebah et al. [37]	✓	X	X
Artifcial ecosystem optimization operators with the WOA	Mostafa et al.[34]	X	X	X
Multi-strategy assisted multi-objective WOA	Yang et al. [32]	X	X	X

However, these methods may still fail to remove complex redundant features because of their inability to communicate directly with classification outcomes. Furthermore, researchers have explored the use of linear dimensionality reduction techniques for feature selection, such as singular value decomposition (SVD) [45] and linear discriminant analysis (LDA) [45]. These linear techniques have shown certain effectiveness, yet they have limitations, such as computational difficulties with SVD and poor performance of LDA on non-Gaussian and small-sample datasets. Some scholars have also experimented with nonlinear dimensionality reduction techniques to select and filter high-dimensional features, such as kernel principal component analysis (KPCA) [46] and multidimensional scaling (MDS) [46]. However, the performance of KPCA can vary significantly depending on the chosen kernel, whereas MDS performance is highly susceptible to outliers in the data.

With the advancement of deep learning, feature selection for deep features has increasingly gained attention. Some researchers have integrated classical feature selection algorithms into deep neural networks to filter deep features. Thakkar et al. [47] proposed a deep neural network feature selection method based on the fusion of statistical importance, which compresses and filters features based on the rank derived from the statistical importance of variance, mean, and median differences of deep features. Acikgoz et al. [47] combined the ReliefF method with the feature extraction layer of convolutional neural networks, facilitating the identification and selection of significant features in the subset. Furthermore, learning-based feature selection methods represent another major category of deep feature selection approaches [48]. Channel selection in deep neural networks can be considered a special task of feature selection. Xue et al. [48] introduced an attention-based progressive channel selection in the task of differentiable neural architecture search, effectively identifying key channels. To further increase the accuracy of channel identification, Xue et al. [49] improved the attention mechanism and proposed a dual-attention adaptive weight architecture, which significantly enhanced the recognition of important channels. Despite the promising results of feature selection methods for deep neural networks, their high demand for computational resources may limit the applicability of these approaches in certain scenarios.

In summary, EC-based feature selection algorithms hold significant potential in addressing high-dimensional feature selection challenges. WOA, as one of the recent representative algorithms in the EC domain, has also received considerable attention in the field of feature selection. However, current studies predominantly focus on single-swarm strategies, which may limit performance in the face of complex feature spaces presented by high-dimensional datasets [50]. Although some studies have recognized the advantages of multi-swarm algorithms in addressing complex optimization problems, introducing multi-population structures to high-dimensional feature selection challenges, the fixed number of sub-populations in existing multi-swarm structures hinders synergistic interactions among individuals, necessitating deeper research into multi-swarm construction mechanisms. Moreover, the feature space transformation function in the existing high-dimensional feature selection algorithms mostly adopts random thresholds, which has the problem of invalid probability flipping, which may lead to the loss of effective features. To overcome these challenges, we propose the EMSWOA algorithm, which, by designing a dynamic multi-swarm mechanism and elite tuning strategy, aims to enhance the feature selection capabilities in high-dimensional feature selection problems, offering a new and effective solution in this field.

3. The proposed method

To improve the quality of feature selection, we propose an EMSWOA. Specifically, the general framework of the EMSWOA is presented in 3.1. The dynamic multi-swarm mechanism based on the centroid distance is described in detail in subsection 3.2. In 3.3, the elite tuning strategy is described, and the fitness function is described in 3.4.

3.1. The framework of the EMSWOA

On one hand, we provide an example of feature selection as illustrated in Fig. 1. First, a set of candidate solutions is initialized and mapped to a binary feature space via a transformation function. Second, this initialized feature subset matrix is entered into the EMSWOA, which undergoes iterative updates and quality assessments to search continuously for the optimal feature subset. Finally, upon completion of the iterative optimization, the optimal feature subset is output.

On the other hand, to provide a more detailed illustration of the EMSWOA feature selection process, we have redrawn the flowchart of the EMSWOA, with Fig. 2 showing the detailed steps of the optimization process. Additionally, for a more intuitive understanding, this paper also presents detailed pseudocode for two new strategies.

First, in the initialization phase, a set of EMSWOA candidate solutions consisting of 0,1 binary codes are initialized, with 0 being not selected and 1 being selected. The formula is as follows:

$$\mathbf{X} = \begin{pmatrix} X_{1,1} & X_{1,2} & \ldots & X_{1,\dim} \\ X_{2,1} & X_{2,2} & \ldots & X_{2,\dim} \\ \ldots & \ldots & \ldots & \ldots \\ X_{N,1} & X_{N,2} & \ldots & X_{N,\dim} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \ldots & 1 \\ 0 & 1 & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots \\ 1 & 1 & \ldots & 0 \end{pmatrix} \tag{12}$$

Second, the subpopulations of the EMSWOA are dynamically constructed through the proposed dynamic multi-swarm mechanism, which is based on the centroid distance. The topology between subpopulations of dynamic multi-swarm is dynamic and reorganized. Compared with the static topology, the dynamic topological multiple population structure can make the interaction of individual information in the search space more diverse and convenient, which facilitates the exchange of merit-seeking information among subpopulations through population reorganization, thus enhancing the diversity of populations in the algorithm space, slowing down convergence and avoiding premature convergence.

Based on the dynamic multiple populations constructed, the individuals in the population are iteratively updated via the encircling prey phase, bubble attack phase and random search phase, and the candidate solutions are discretely mapped in binary via the transformation function, and the following model [3] is used for discrete mapping in this paper:

$$f(x) = \frac{1}{1 + e^{-2x}} \tag{13}$$

$$X_{i,j} = \begin{cases} 1 & , if\ r < \mathrm{f}\left(\Delta X_{i,j}\right) \\ 0 & , if\ r \geq f\left(\Delta X_{i,j}\right) \end{cases} \tag{14}$$

where r is a normally distributed stochastic number with values in [0,1], $f(x)$ is a discrete mapping function to complete the binary conversion, and $\Delta X_{i,j}$ denotes the amount of change in the i-th dimension of the j-th candidate solution.

Finally, the proposed elite tuning strategy is used to correct for invalid probability flips in each iteration of the EMSWOA, improving the quality of subset selection and the velocity of the EMSWOA approximation towards the optimal solution. By continuously cycling the above global optimization process, the best subset is thus obtained.

3.2. Dynamic multi-swarm mechanism based on centroid distance

The global search performance of metaheuristic algorithms is influenced by their local exploitation and global exploration abilities. In multiple swarm evolutionary algorithms, local information exploitation is achieved through intra-subpopulation search, and global information exploration is achieved through interactive recombination between subpopulations. However, when performing a search within a subpopulation, the size of the subpopulation's search space and the number of individuals within it have a significantly influence over the information exploited in that space; specifically, the larger the search space is, the

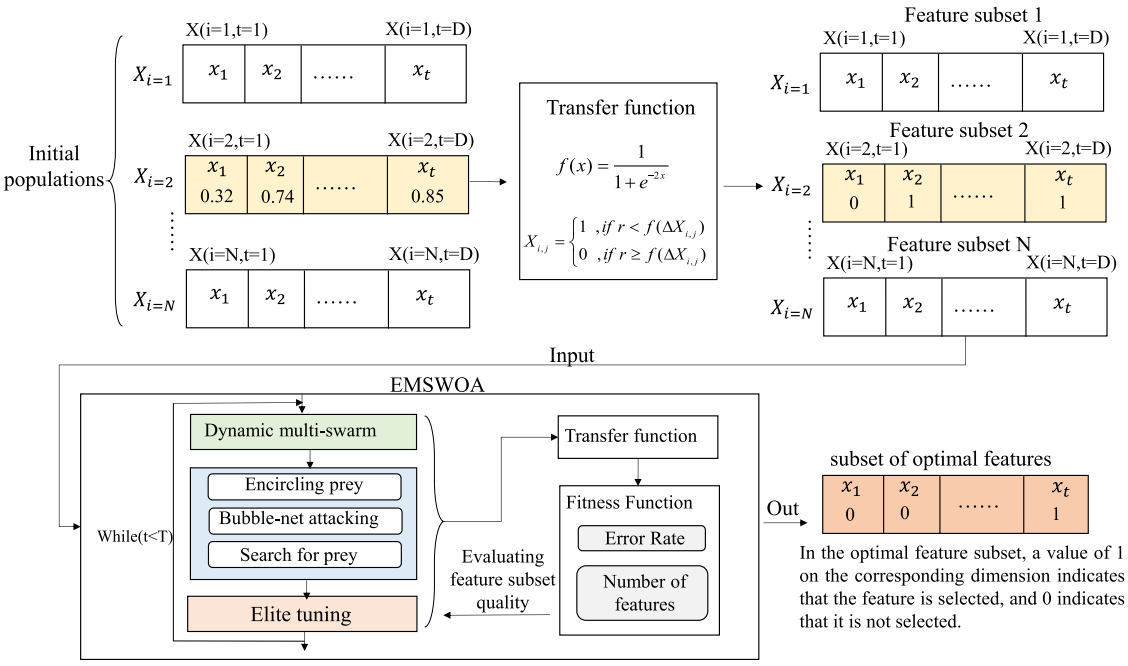


Fig. 1. Example of an EMSWOA-based feature selection process.

greater the number of individuals included and the more favorable the algorithm's search for excellence. To avoid the problem of inadequate use of local information due to the small number of individuals contained in populations with a larger search space when conducting intra-subpopulation search, and to enhance the ability to explore the local space, we propose a dynamic multi-swarm mechanism by centroid distance construction, as shown in Fig. 3.

In the EMSWOA, the dynamic mechanism of dynamic multiple populations can be divided into two parts. One part refers to the dynamic process of changing the number of subpopulations after each reorganization cycle. The other part refers to the dynamic recombination process between subpopulations. To clearly describe the dynamic multi-swarm mechanism based on the centroid distance, we divide it into four steps: the dynamic generation of the number of sub-populations, the construction process of sub-populations, the search mechanism based on the local sparsity metric, and the modification of the search process of the WOA. The pseudocode of the dynamic multi-swarm mechanism based on the centroid distance is shown in Algorithm 1.

Step1: Dynamic generation of the number of subpopulations.

The number of subpopulations to be generated at each population reorganisation throughout the merit search cycle is generated according to the following model:

$$\begin{aligned} groups(\mathrm{g}) &= sort(round(group_{\min} + rand \\ &\quad * (group_{\max} - group_{\min}))) \\ s.t. &\quad 0 < g < round(T/R) \end{aligned} \tag{15}$$

where$round(x)$denotes a rounding function, $rand(x)$ denotes a normally distributed random number, $group_{\max}$ and $group_{\min}$ denote the maximum and minimum populations, respectively, which are defined as 20 and 1, and g indicates the population size of the subpopulation regroups. T indicates the total number of iterations, which is 100 in this paper. R is referred to as the reorganization cycle, where the population reorganizes after every R iteration, and is 10 in this paper. $round(T/R)$denotes the number of times the total population must to be reorganized within T iterations; in this paper, it is 10, i.e., within 100 iteration cycles, 10 population reorganizations must to be experienced, and all the sub-populations are combined into one population at the last population reorganization. $sort(x)$ is a sorting function that is sorted in descending order. Its purpose is to gradually reduce the number of subpopulations with each iteration, as having a larger number of subpopulations during the initial period of evolution can enhance the WOA's ability to exploit local information. In the last reorganization cycle, all the subpopulations are reduced to the same population, which helps to enhance the connectivity between multiple population topologies, boosts the global exploration capability and facilitates enhanced learning of interaction information between subpopulations, thus helping to achieve the best subset.

Step2: Process of constructing the subpopulations.

When constructing the subpopulations for the current recombination cycle, the number of base whales contained in each subpopulation is first initialized and the number of base whales gnum_{init}contained in each subpopulation is calculated from the following model:

$$\mathrm{gnum}_{init} = floor\left(\frac{N}{groups(\mathrm{g})}\right) - 1 \tag{16}$$

where N denotes the number of all whales and $floor(x)$ denotes rounding down. Considering the sparsity in the subpopulations, the algorithm adds more search agents for subpopulations with larger search spaces in order to enhance the spatial searching abilities of subpopulations with larger search spaces. After the initial subpopulation is constructed, the distance between an individual and the centroid of the subpopulation is calculated for each individual in the subpopulation via the centroid distance measurement mechanism to obtain approximate moments of the size of the subpopulation space, and the mathematical model is shown below.

$$dist_k = \max\left(\sqrt{X_k(t) - \sum_{1}^{n} X_k(n) \Big/ n)^2}\right) \tag{17}$$

where k indicates the subpopulation number and $X_{k,n}$ denotes the n-th individual whale belonging to the k-th subpopulation. After initializing the subpopulations, the optimization-seeking capability of the algorithm in the sparse space is enhanced by adding more individuals to the sub-populations with larger $dist_k$ values. The remaining individuals are added one by one to the subpopulations that need to be added in this round based on the size of $dist_k$; thus, more search agents obtain a more

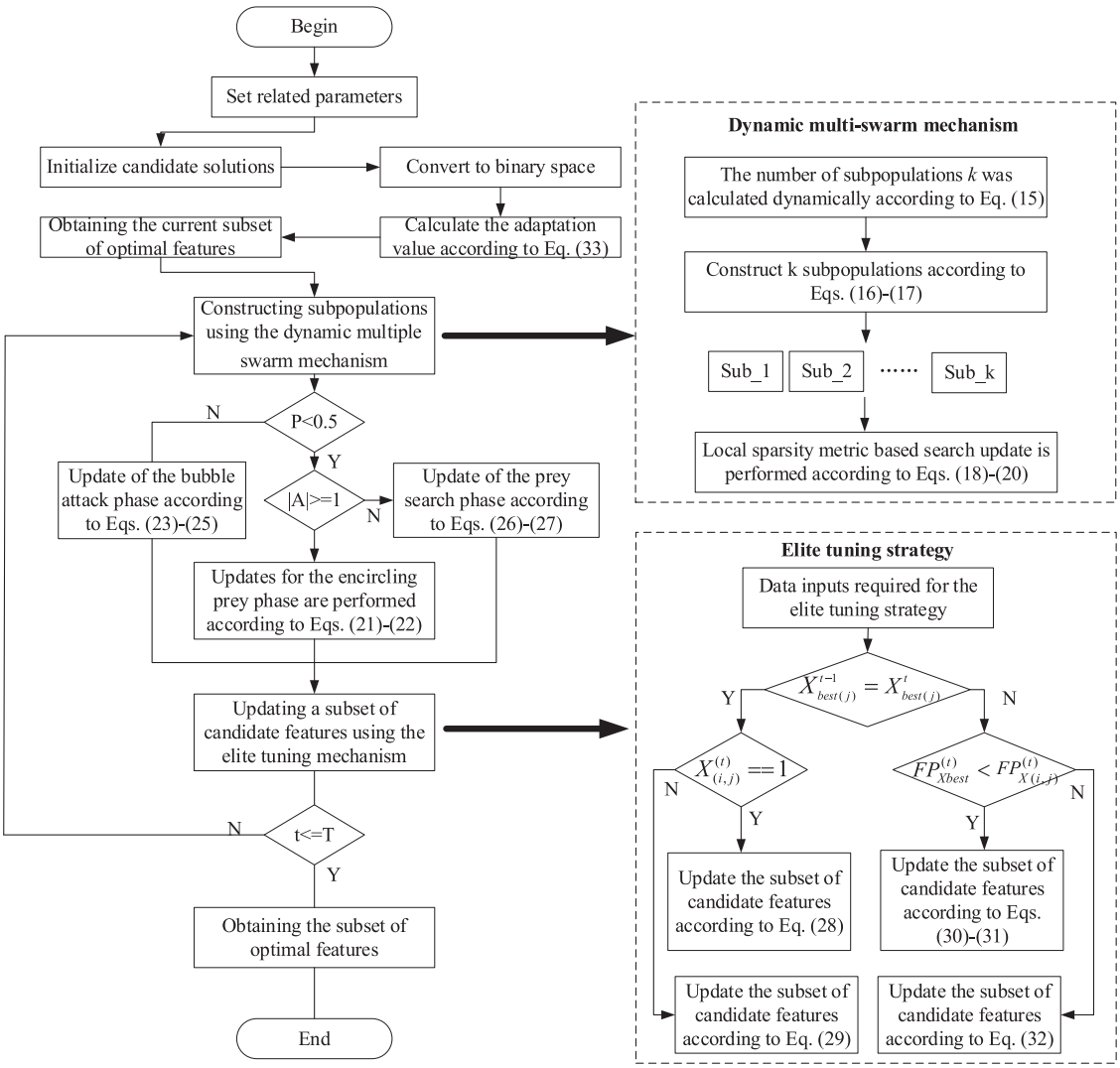


Fig. 2. Overall flowchart for feature selection via the EMSWOA.

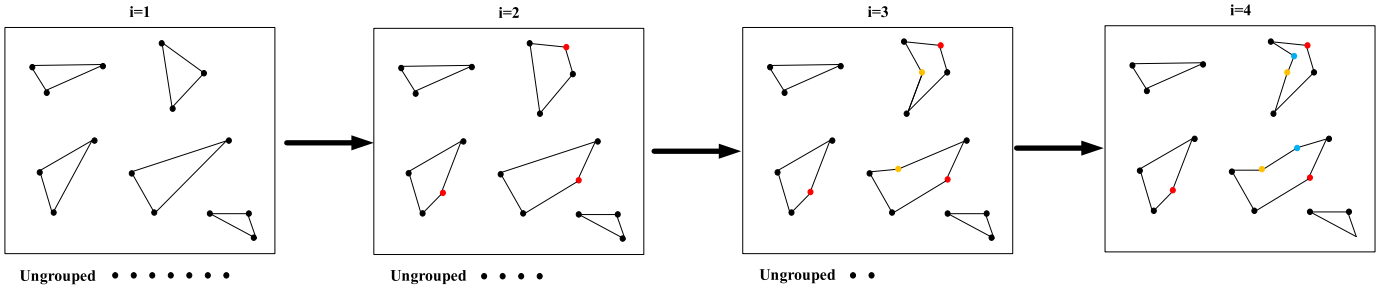


Fig. 3. Example of multiple swarm regrouping based on the centroid distance metric.

discrete the decision space. An iterative search for superiority with period R is subsequently performed in the respective new search subspace. By performing one cycle of learning in each subpopulation before reorganization and information exchange, it is possible to avoid the loss of local information caused by frequent reorganization of the algorithm when the subpopulation itself does not learn sufficiently.

Step3: Search mechanism that is based on locally sparse metrics.

Multi-swarm algorithms enhance the exploitation and exploration capabilities by dividing the population into multiple subpopulations. However, as each subpopulation contains far fewer candidate solutions than a single-swarm algorithm, multi-swarm algorithms are more sensitive to the diversity of candidate solutions within subpopulations. To maintain diversity among the candidate solutions in the subpopulations, we introduce a local sparsity metric mechanism (LSD) [51] to guide individuals towards those with greater LSD, thus preserving population diversity. For clarity, this paper provides a schematic of the distance intervals based on fitness values, as shown in Fig. 4.

The LSD represents the local sparsity of an individual, which is determined by the individual's degree of congestion $con(i)$ and dispersion $dis(i)$. $con(i)$ focuses on global information within the subpopulation, whereas $dis(i)$ measures the neighbouring information of individual X_i. The LSD considers the crowding of individual X_i from both

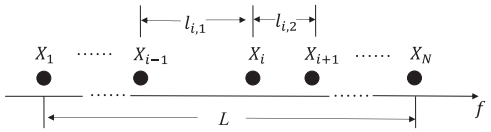


Fig. 4. Example calculations for $l_{i,1}$,$l_{i,2}$and$L$.

global and local perspectives. Specifically,$con(i)$ calculates the proportion of the neighbouring interval of X_i to the total space, and represents the individual's congestion level. $dis(i)$ is used to calculate the dispersion between individual X_i and neighbouring individuals, assessing whether the neighbouring individuals are more uniformly distributed; the less uniform the distribution is, the smaller the value of $dis(i)$. Moreover, to reduce the complexity of calculating the LSD, $con(i)$ and $dis(i)$ are normalized, as illustrated in the mathematical model [51] below.

$$con(i) = \frac{l_{i,1} + l_{i,2}}{L} \tag{18}$$

$$dis(i) = \frac{\min(l_{i,1}, l_{i,2})}{max(l_{i,1}, l_{i,2})} \tag{19}$$

$$LSD(i) = \frac{con(i)}{\max(con)} \frac{dis(i)}{\max(dis)} \tag{20}$$

$l_{i,1}$ and $l_{i,2}$ represent the fitness interval distances between X_i and its two nearest neighbours, and L is the maximum interval among all individuals within the subpopulation spaces. When calculating the dispersion of neighbouring individuals via LSD, the current best and worst values are excluded, setting the LSD values of these two individuals to zero. This is because the best and worst individuals, being at the extremes of fitness evaluations, each have only one neighbour, making it impossible to calculate their LSD.

The EMSWOA enables individuals to transition their learning from a crowded space to a discrete space by guiding them to approach individuals with larger LSD values, thus maintaining population diversity. To prevent precocious convergence, based on the suggestion in the literature [51], individuals with larger LSD are directly passed on to the next generation. The computational procedure is detailed as follows. First, the LSD corresponding to each candidate solution is computed. Second, the candidate solutions are sorted in ascending order based on the value of the LSD, and the rank of the candidate solutions in the population. Third, if the LSD value corresponding to a candidate solution is less than or equal to $N * rand$, the candidate solution is added to X_{sub}, which is used for population updating of candidate solutions with smaller LSD values. Fourth, from subpopulation k, based on the ascending sequence of LSD, a candidate solution $X^i_{k,div}$ with an LSD value greater than $LSD(i)$ is randomly selected. Finally, based on the fitness values of the individuals in subpopulation k, the optimal solution $X^i_{k,best}$ with the smallest fitness value in the current subpopulation is calculated and obtained.

Step 4: Improve the WOA based on dynamically constructed subpopulations and a local sparse metric mechanism.

In the EMSWOA, we use LSD to filter out the individuals in the subpopulation that need to be updated X_{sub}. First, based on the dynamic multiple population grouping, the current optimal solution $X^i_{k,best}$ in subpopulation k is used to replace the global optimal solution $X^*(t)$ of a single population in the original algorithm to realize the local detection capability within the subpopulation. Second, X_{rand} in the original algorithm is replaced by using the random candidate solution $X^i_{k,div}$ that is larger than $LSD(i)$ in subpopulation k to realize the candidate solution

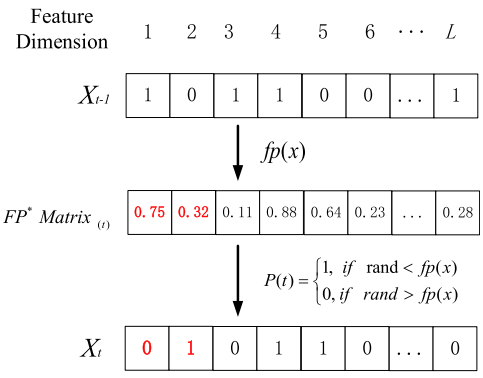


Fig. 5. Example of an invalid probability flip for discrete space.

detection update based on the LSD metric. Finally, based on the above analysis, the stages of the WOA algorithm are modified, and the improved search stages are represented as follows:

(1) Encircling prey

$$D = \left| C \cdot X^i_{k,best}(t) - X^i_k(t) \right| \tag{21}$$

$$X^i_k\left(t+1\right) = X^i_{k,best}(t) - A \cdot D \tag{22}$$

(2) Bubble-net attack method

$$X^i_k\left(t+1\right) = D' \cdot e^{bl} \cdot \cos\left(2\pi l\right) + X^i_{k,best}(t) \tag{23}$$

$$D' = \left| X^i_{k,best}(t) - X^i_k(t) \right| \tag{24}$$

$$X^i_k\left(t+1\right) = \begin{cases} X^i_{k,best}(t) - A \cdot D & if \;\; p < 0.5 \\ D' \cdot e^{bl} \cdot \cos\left(2\pi l\right) + X^i_{k,best}(t) & if \;\; p \geq 0.5 \end{cases} \tag{25}$$

(3) Search for prey

$$X^i_k\left(t+1\right) = X^i_{k,div}(t) - A \cdot D \tag{26}$$

$$D = \left| C \cdot X^i_{k,div}(t) - X^i_k(t) \right| \tag{27}$$

Among them, the other parameters, except $X^i_{k,div}$ and $X^i_{k,best}$, are consistent with the meaning and parameter setting in the WOA, and the description is not repeated there.

Compared to the single-swarm WOA algorithm, the multi-swarm based EMSWOA exhibits stronger optimization capabilities. Firstly, the multi-swarm structure allows parallel exploration in different search areas of the feature space through distinct subpopulations, enhancing coverage within the solution space and preventing the omission of more discriminative feature subsets. Secondly, the structure facilitates the discovery of global optima by periodically interacting and sharing information between different populations, effectively preventing the algorithm from stagnating. Additionally, the multi-swarm structure helps maintain diversity among candidate solutions in the population, by allowing different subpopulations to evolve independently in various areas, thereby reducing the risk of overfitting [52]. Lastly, the multi-population structure aids in maintaining a balance between exploitation and exploration, thus enhancing the algorithm's global search capabilities [53].

Algorithm 1. The pseudocode of dynamic multi-swarm mechanism based on centroid distance

Input: The population size N, the maximum iteration period T, the fitness value *fitness*, the population reconstruction period R, and the associated parameters $A, C, b, l \text{ and } p$ of WOA.

Output: Global optimal solution X_{best}

while $t < T$ **do**

$groups \leftarrow$ The number of subpopulations was calculated according to Eq. (15).

if $t=Tg$ then

$gnum_k^{init} \leftarrow$ The initial individuals contained in the subpopulation were randomly generated according to Eq. (16).

$gnum_k \leftarrow$ On the basis of $gnum_k^{init}$, the remaining individuals are added to the subpopulation based on the center distance metric of Equation (17).

Tg=Tg+R; The subpopulation is reconstructed when the number of iterations is every R cycles apart.

end

for i=1:N do

$LSD(i) \leftarrow$ Calculate the local sparsity metric according to Eq. (18)-(20).

end

$ranking \leftarrow sort(LSD)$, the calculated LSD were sorted in ascending order.

for i=1 : N do

if $randking(i) \le N * rand$ then

Deposit into X_{sub} the individuals in the subpopulation that need to be updated.

$X_{k,best}^i \leftarrow$ Obtain the optimal solution corresponding to the ith individual from subpopulation k.

$X_{k,div}^i \leftarrow$ From subpopulation k, a candidate solution with a value of LSD greater than $LSD(i)$ is randomly selected based on the ascending sequence of LSD.

end

for i=1:N **do**

if $X_k^i \in X_{sub}$ **then**

Calculate and update $A, C, b, l \text{ and } p$

if $p < 0.5$ **then**

if $|A| < 1$ **then**

The update of the encircling prey phase was carried out according to Eq. (21)-(22).

else if $|A| \ge 1$ **then**

A random X_i is chosen and the position is updated according to Eq. (23)-(25).

end if

else if $p \ge 0.5$ **then**

Position update of the spiral surround according to Eq. (26)-(27).

end if

Boundary checks for $X(t)$

else

X_k^i is directly inherited to the next generation.

end

end for

if $X(t)$ is better, update to X_{best}

$t = t + 1$

end while

return X_{best}

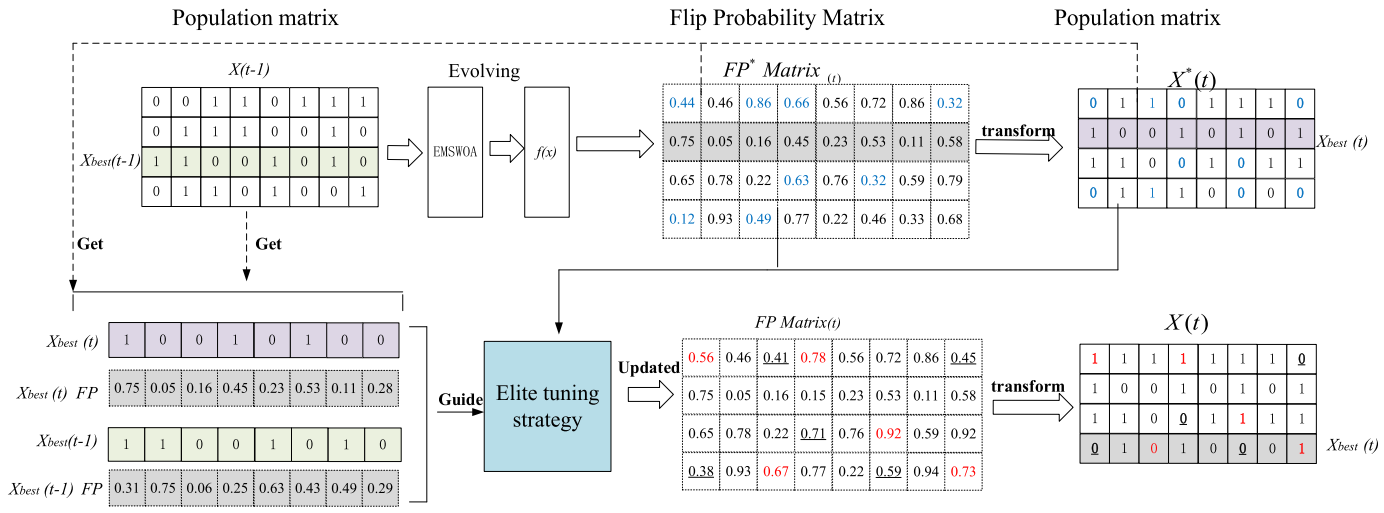


Fig. 6. Elite tuning strategy process.

Overall, the dynamic multi-swarm mechanism based on the centroid distance can effectively jump out of the local optimum and avoid premature convergence. Specifically, the population is divided into several subpopulations through the centre distance-based metric, and in the early iteration, the sub-populations can separately search in the high-dimensional feature selection space individually. As the iteration continues, the constant population reorganization enhances the topological connectivity among the populations, which facilitates the search of the complex space and realizes the full exploitation and exploration of the complex solution space such that the algorithm effectively jumps out of the local optimal solution and effectively avoids precocious convergence.

3.3. Elite tuning strategy

Inspired by the literature [3], this paper employs the commonly used S-type function in this field as the transfer function to achieve conversion to binary space, as shown in Eqs. (13)-(14). The elite tuning strategy proposed in this paper optimizes the S-type transfer function. Specifically, the S-type (or V-type) transfer functions convert real-valued solutions to probability values between 0 and 1 during binary transformation. A random value within the range of 0–1 is then used as a threshold. By comparing the flip probability with this random threshold, it determines whether a feature is selected (the value is 1 when it is selected). However, due to the randomness of the threshold, which varies across dimensions, this can lead to ineffective flips, as illustrated in Fig. 5. For instance, when the flip probability Fp_i^1 is 0.75 and the random threshold r is 0.8, the feature is not selected (indicated by a value of 0) because $Fp_i^1 < r$. Conversely, when the flip probability Fp_i^2 is 0.32 and the random threshold r is 0.3, the feature is selected (indicated by a value of 1) because $Fp_i^1 > r$. Due to the randomness of thresholds across different dimensions, even though $Fp_i^1 > Fp_i^2$, X_i^1 is not selected (indicated by$X_i^1 = 0$), leading to the issue of ineffective probability flips.

The elite tuning strategy uses the elite candidate solutions in the algorithm space as a benchmark and guides the correction of the probability flip matrix and the eigenvalues of the current population obtained after the mapping of the transformation function. The selection of elite individuals is critical to the elite tuning strategy, and it directly influences the subsequent correction benchmarks. To ensure the accuracy of the assessment of feature importance among elite individuals, two consecutive generations of elite individuals were selected as the benchmark for elite correction in this study. The process of the elite tuning strategy is shown in Fig. 6. By employing the correction rules defined in Eqs. (28)–(32), this strategy increases the probability values of important features or decreases the thresholds to increase the likelihood of feature selection.

The process of the elite tuning strategy relies on the elite individuals of the two generations before and after making dimension-by-dimension corrections to the candidate solutions. To clearly describe the correction process, we divide the process into three steps: data input, two generations of elite consistency, and two generations of elite inconsistency. To make this easier to understand, we present pseudocode for the elite tuning strategy, as shown in Algorithm 2.

Step 1: Data inputs required for the elite tuning strategy. The optimal candidate solution (elite individual) $X_{best(j)}^t$ and the feature flip probability $FP_{Xbest}^{(t)}$ corresponding to the elite individual are obtained from the EMSWOA algorithm space at round t iteration, as are the optimal candidate solution $X_{best(j)}^{(t-1)}$ and the feature flip probability $FP_{Xbest}^{(t-1)}$ corresponding to that elite individual at the previous round iteration (t-1). The candidate solutions in the EMSWOA algorithm space are then corrected dimension by dimension based on the two generations of elite individuals and their characteristic flip probabilities.

Step 2: Two generations of elite consistency. When $X_{best(j)}^t$ equals $X_{best(j)}^{(t-1)}$, this indicates a high degree of agreement between the two generations of elite solutions and high accuracy in the assessment of the importance of the features corresponding to the optimal solution in that dimension. At this point, if the candidate solution does not agree with the elite solution, it can be corrected via the following process.

(1) When a feature in a corresponding dimension of the elite solution vector $X_{best(j)}^{(t)}$ is selected (i.e., the feature is marked as 1), and the feature in the corresponding dimension of the candidate vector $X_{(i,j)}^{(t)}$ to be corrected is marked as 0, this indicates an inaccurate assessment of feature importance for the candidate solution vector $X_{(i,j)}^{(t)}$ in that dimension. At this point, forcibly setting the vector $X_{(i,j)}^{(t)}$ to 1 would cause the candidate solution to lose its ability to explore unknown spaces. Therefore, this

Table 2
Details of the high-dimensional feature selection dataset.

No.	Datasets	#Features	#Instances	#Classes
1	11_Tumor	12533	174	11
2	9_Tumor	5726	60	9
3	Brain_Tumor_1	5920	90	5
4	Brain_Tumor_2	10367	50	4
5	DLBCL	5469	77	2
6	Leukemia_1	5327	72	3
7	Leukemia_2	11225	72	3
8	Leukemia_3	11225	72	3
9	Lung_Cancer	12600	203	5
10	Prostate_Tumor_1	5966	102	2
11	GLI_85	22283	85	2
12	SMK-CAN−187	19993	187	2
13	ALLAML	7129	72	2
14	Colon	2000	62	2
15	Lymphoma	4026	96	9
16	Nci9	9712	60	9
17	Leukemia	7070	72	2
18	Prostate-GE	5966	102	2
19	Yale	1024	165	15

Table 3
Parameter settings of related algorithms.

Algorithm	Parameter
GOOSE	$coe < 0.17$, N = 100, T = 100
CSOFS	N = 100, T = 100
BCCAS	$AP = 0.1$,$fl = 2$, N = 100, T = 100
HCLDMS-PSO	$a = 0.05$,$b = 0.45$,$CL \in [0.99, 0.2]$, $C_1 \in [2.5, 0.5]$,$C_2 \in [0.5, 0.25]$, N = 100, T = 100
MWOA	$a \in [2, 0]$,$b = 1$,$l \in [-1, 1]$,$\beta \in [0, 1]$, N = 100, T = 100
HWOAG	$a \in [2, 0]$,$b = 1$,$l \in [-1, 1]$, N = 100, T = 100
EMSWOA	$a \in [2, 0]$,$b = 1$,$l \in [-1, 1]$, $group_{\max} = 20$,$group_{\min} = 1$,R= 10, $\alpha = 0.45$,$\beta = 0.2$,$\lambda = 0.5$,N = 100, T = 100

study employs a normal random variable $r(x)$ for random perturbation, and sets αas a threshold to determine whether a feature should be selected. We set α to 0.45 in order for $X_{(i,j)}^{(t)}$ to have more than a 50 % probability of being selected based on that dimensional feature. The process can be represented by the following model.

$$X_{(i,j)}^{(t)} = r(x) > \alpha, \qquad if\ X_{best(j)}^{(t)} == 1 \quad and \quad X_{(i,j)}^{(t)} == 0 \tag{28}$$

(2) When $X_{best(j)}^{(t)}$ has a feature selection score of 0 in that dimension, but $X_{(i,j)}^{(t)}$ is not 0. The sum of the number of features ($\#Seleted$) contained in the currently selected subset and the proportion of all features ($\#All$), via the threshold β (set to 0.2), is used as a dynamic threshold that jointly controls the correction in this dimension. When the selected candidate subset contains more features, the dynamic threshold is larger, thus achieving better feature redundancy.

$$\begin{aligned} &X_{(i,j)}^{(t)} = r(x) > \left(\beta + \left(\frac{\#Seleted}{\#All}\right)\right) \\ &s.t.\ \ if\ X_{best(j)}^{(t)} == 0 \quad and \quad X_{(i,j)}^{(t)} \neq 0 \end{aligned} \tag{29}$$

Step 3: Two generations of elite inconsistency. When $X_{best(j)}^{(t)}$ is not equal to $X_{best(j)}^{(t-1)}$, $X_{best(j)}^{(t)}$ is used as the dominant party for the elite tuning strategy. To avoid a reduction in the diversity of candidate feature subsets, we use only the dimension with a value of 1 in $X_{best(j)}^{(t)}$ as the dimension to be corrected for the other candidate subsets of the feature space.

(1) When the value of $FP_{Xbest}^{(t)}$ is less than the flip probability of the candidate individual to be corrected, but the $X_{best(j)}^{(t)}$ is 1, an invalid flip in the feature evaluation of $FP_{X(i,j)}^{(t)}$ has occurred. Then the value of $FP_{X(i,j)}^{(t)}$ is increased by using the flip probability of the elite individual of two generations to improve the chance of being selected for the feature in this dimension. The specific calculation formula is as follows:

$$FP_{X(i,j)}^{(t)} = FP_{X(i,j)}^{(t)} + r(x) * \left|\left(FP_{best(j)}^{(t)} - FP_{best(j)}^{(t-1)}\right)\right| \tag{30}$$

$$X_{(i,j)}^{(t)} = \begin{cases} 1, & if \quad FP_{X(i,j)}^{(t)} > \lambda \\ 0, & if \quad FP_{X(i,j)}^{(t)} \leq \lambda \end{cases} \tag{31}$$

where $r(x) \in (0, 1)$ is normally distributed and λ is the threshold for whether a feature is selected, which in this paper is 0.5. Using the amount of variation in the flip probability of elite individuals over two generations and random perturbations, $FP_{X(i,j)}^{(t)}$ is updated such that features in that dimension boost the possibility of being chosen.

(2) When the value of $FP_{Xbest}^{(t)}$ is greater than the flip probability $FP_{X(i,j)}^{(t)}$ of the current candidate individual to be corrected. At this point, a value of 1 for the corresponding dimension of $X_{best(j)}^{(t)}$ and 0 for $X_{(i,j)}^{(t)}$ indicates a case of inaccurate feature importance assessment due to insufficient quality of the subset of features of the candidate individual $X_{(i,j)}^{(t)}$ to be corrected, independent of the flip probability. To increase the global exploration ability of $X_{(i,j)}^{(t)}$, we guide a subset of candidate features towards a better feature space by elite values and update the mathematical formula as follows:

$$\begin{aligned} X_{(i,j)}^{(t)} = X_{(i,j)}^{(t)} + \mathrm{r}(x) \times \left(\left(1 - \frac{t}{T}\right)\right. \\ \left.\times \frac{1}{1 + e^{FP_{Xbest(j)}^{(t)} - FP_{X(i,j)}^{(t)}}}\right) * \left|X_{best(j)}^{(t)} - X_{(i,j)}^{(t)}\right| \end{aligned} \tag{32}$$

Algorithm 2. The pseudocode of elite tuning strategy

Input: $X_{best}^{(t-1)}, FP_{Xbest}^{(t-1)}, X_{best}^{(t)}, FP_{Xbest}^{(t)}, X^{(t)}$, Dim

Output: Global optimal solution X_{best}

while $t < T$ **do**
  for i=1:N **do**
    for j=1:Dim do
      if $X_{best(j)}^{(t-1)} = X_{best(j)}^{(t)}$ **then**
        if $X_{best(j)}^{(t)} = 1 \,\&\, X_{i,j}^{(t)} = 0$ **then**
          Correct $X_{i,j}^{(t)}$ according to Eq. (28).
        else if $X_{best(j)}^{(t)} = 0 \,\&\, X_{i,j}^{(t)} \neq 0$ then
          Correct $X_{i,j}^{(t)}$ according to Eq. (29).
        end if
      else if $X_{best(j)}^{(t-1)} \neq X_{best(j)}^{(t)} \,\&\, X_{best(j)}^{(t)} = 1 \,\&\, X_{(i,j)}^{(t)} = 0$
        if $FP_{Xbest}^{(t)} < FP_{X(i,j)}^{(t)}$ then
          Correct $X_{i,j}^{(t)}$ according to Eq. (30) and Eq. (31).
        Else
          Correct $X_{i,j}^{(t)}$ according to Eq. (32).
        End if
      End if
      Boundary checks for X
    end for
  end for
  if X* is better, update to X_{best}
  $t = t + 1$
end while
return X_{best}

Overall, the elite tuning strategy corrects for invalid probability flips and accelerates algorithm convergence. This improvement is attributed to the elite tuning strategy, which enhances the quality of the subset of candidate features. The strategy uses elite individuals within the population as a benchmark. By assessing the importance of features in different dimensions, the elite solution is compared with the candidate solution. This comparison, combined with the flipping probability, allows for the correction of features with invalid flips. As a result, the quality of the subset of candidate features is improved. This approach prevents ineffective individual searches and accelerates the convergence process of the EMSWOA.

3.4. Fitness function

The fitness evaluation function for FS is a method used to assess the quality of features. In the FS task, it is desirable to use the smallest subset of features to maximize classification accuracy. Therefore, the fitness function is also coupled by two metrics, the number of feature subsets and the accuracy,the mathematical model for which is shown below [3]:

$$fitness = \alpha * ErrorRate + \left(1 - \alpha\right) * \frac{\#Selected}{\#All} \tag{33}$$

where *ErrorRate* indicates the error rate, *#Selected* indicates the number of features selected, and *#All* denotes all features; it is used to control the

Table 4
Comparison of the average classification accuracy (%) and standard deviation of the EMSWOA and SOTA.

Dataset	GOOSE	CSOFS	BCCAS	HCLDMS-PSO	MWOA	HWOAG	EMSWOA
11_Tumor	87.47 ± 0.26	88.51 ± 0.70	88.97 ± 1.31	88.05 ± 1.59	90.23 ± 1.15	88.39 ± 0.63	**92.64 ± 1.37**
9_Tumor	51.00 ± 4.80	61.00 ± 1.49	64.33 ± 0.91	63.67 ± 3.80	69.33 ± 2.53	62.00 ± 2.17	**72.33 ± 4.35**
Brain_Tumor_1	86.22 ± 1.27	89.11 ± 0.50	88.67 ± 0.50	88.67 ± 1.45	91.33 ± 2.28	89.11 ± 0.93	**94.89 ± 1.27**
Brain_Tumor_2	84.00 ± 1.41	87.20 ± 1.79	84.40 ± 0.89	84.40 ± 1.67	87.20 ± 1.79	84.00 ± 0.00	**92.40 ± 1.67**
DLBCL	95.84 ± 0.58	96.88 ± 0.71	95.32 ± 0.71	94.29 ± 0.71	98.44 ± 1.09	95.58 ± 0.71	**100.00 ± 0.00**
Leukemia_1	95.28 ± 1.24	95.56 ± 0.62	97.22 ± 0.98	95.28 ± 0.76	98.89 ± 1.16	96.67 ± 0.76	**99.44 ± 0.76**
Leukemia_2	90.83 ± 0.76	92.22 ± 1.24	91.67 ± 0.00	91.94 ± 0.62	95.28 ± 0.76	91.39 ± 0.62	**97.50 ± 1.16**
Leukemia_3	98.06 ± 0.76	98.33 ± 0.62	97.78 ± 0.76	97.22 ± 0.00	98.89 ± 1.16	97.50 ± 0.62	**100.00 ± 0.00**
Lung_Cancer	95.47 ± 0.54	95.86 ± 1.24	95.76 ± 0.89	95.27 ± 0.27	97.24 ± 0.56	95.37 ± 0.44	**98.03 ± 0.35**
Prostate_Tumor_1	90.98 ± 1.07	91.37 ± 0.44	91.18 ± 0.00	90.59 ± 0.54	91.76 ± 1.12	90.78 ± 0.54	**97.06 ± 0.69**
GLI_85	88.71 ± 1.34	91.76 ± 1.44	90.35 ± 0.53	90.35 ± 0.98	91.76 ± 1.44	90.82 ± 1.53	**96.71 ± 1.74**
SMK-CAN−187	67.70 ± 2.22	72.73 ± 0.65	70.27 ± 0.61	69.30 ± 0.61	71.02 ± 3.10	71.02 ± 0.59	**75.83 ± 1.75**
ALLAML	87.22 ± 4.95	88.89 ± 0.00	93.89 ± 2.52	88.33 ± 0.76	91.94 ± 1.81	90.00 ± 1.16	**100.00 ± 0.00**
Colon	83.87 ± 3.23	86.77 ± 0.72	91.94 ± 1.14	85.48 ± 1.14	90.65 ± 2.10	88.71 ± 1.98	**96.77 ± 1.14**
Lymphoma	95.63 ± 0.47	95.83 ± 0.00	95.63 ± 0.47	94.79 ± 0.00	95.83 ± 0.00	95.63 ± 0.87	**96.04 ± 0.47**
Nci9	53.67 ± 1.83	61.67 ± 1.18	65.33 ± 2.47	58.00 ± 5.06	62.67 ± 1.90	60.33 ± 1.83	**69.00 ± 2.53**
Leukemia	95.00 ± 1.58	93.89 ± 0.76	97.78 ± 0.76	93.61 ± 0.76	97.22 ± 0.98	96.39 ± 0.76	**100.00 ± 0.00**
Prostate-GE	89.22 ± 2.94	91.37 ± 0.44	93.14 ± 1.55	91.18 ± 0.00	91.57 ± 0.88	91.57 ± 0.88	**95.49 ± 1.32**
Yale	56.73 ± 4.19	66.42 ± 0.92	68.12 ± 0.92	62.67 ± 2.52	67.88 ± 1.05	66.91 ± 1.10	**70.18 ± 2.89**

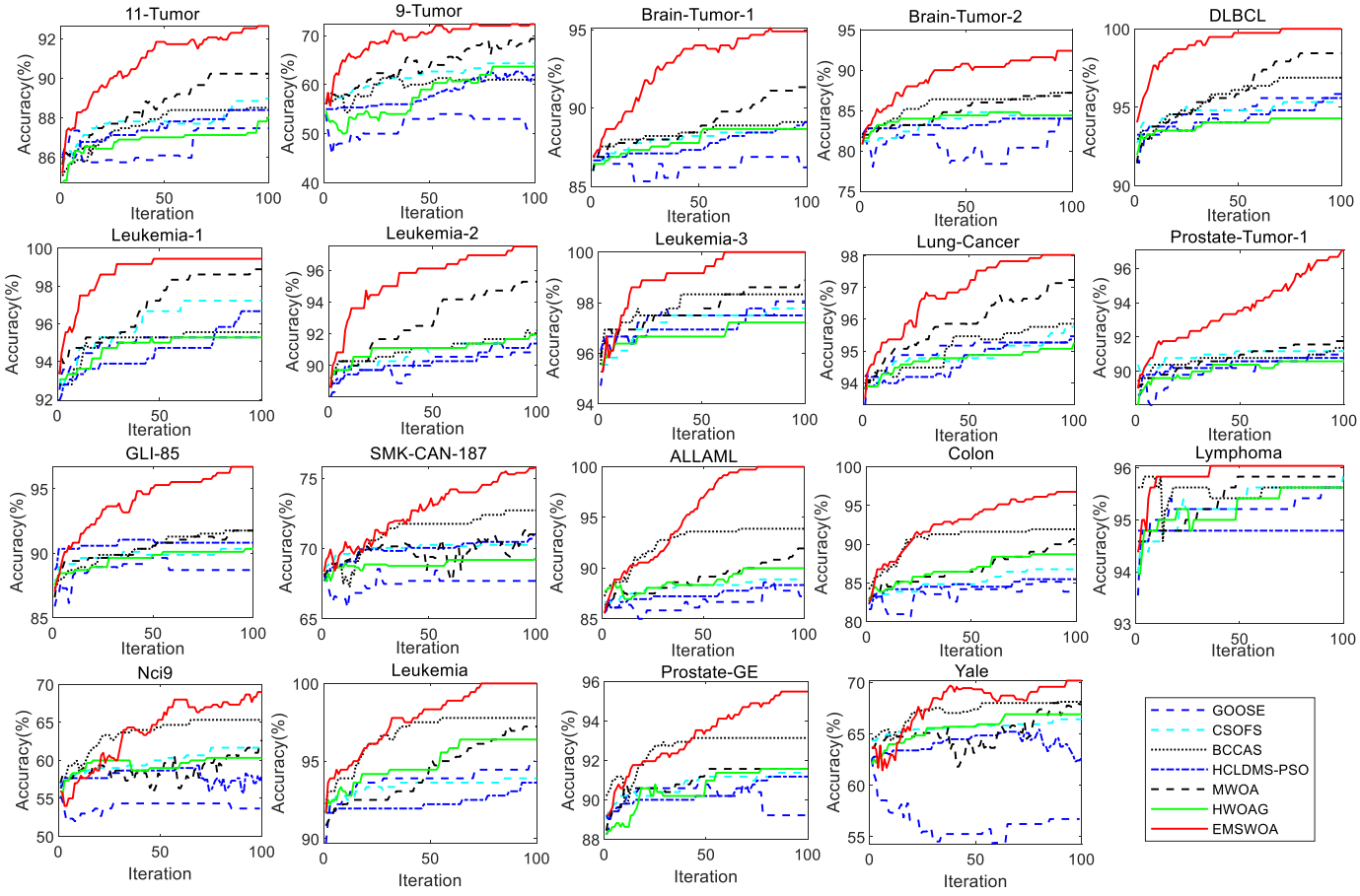


Fig. 7. The average accuracy convergence curves of the EMSWOA with SOTA methods.

balance between the error rate and the number of features, and is set in the literature [23], with α having values in the range [0,1]. To avoid error bias due to sample data, the average error obtained by via 10-fold cross-validation is used to guide the feature selection process of the EMSWOA.

3.5. Time-complexity analysis

Time complexity analysis of swarm intelligence algorithms allows for a more comprehensive assessment of the performance of the algorithms. In this paper, the time complexity analysis of the WOA and EMSWOA, where the time complexity is denoted by O, is performed as follows.

(1) Original WOA

In the original WOA, the time complexity of population initialization

Table 5
Comparison between the EMSWOA and SOTA methods in terms of the average number of selected features.

Dataset	GOOSE	CSOFS	BCCAS	HCLDMS -PSO	MWOA	HWOAG	EMSWOA
11-Tumor	5041.6	3820.2	5328.2	8355.0	3265.6	6546.0	**1097.0**
9-Tumor	532.2	1517.0	2413.4	1790.4	542.8	1244.6	**233.2**
Brain-Tumor−1	284.8	1782.8	2515.0	3753.8	1141.6	2664.6	**122.0**
Brain-Tumor−2	1712.4	3924.4	4459.0	7130.0	2652.0	6261.0	**336.8**
DLBCL	1223.6	1656.4	2361.8	3616.0	919.4	2343.8	**493.2**
Leukemia−1	2973.4	2015.2	2242.4	3435.8	1369.8	2430.8	**506.8**
Leukemia−2	3597.6	2975.2	3059.4	5155.8	1359.0	3817.8	**428.2**
Leukemia−3	3993.4	5083.6	4807.0	7777.2	3124.6	6602.6	**1054.4**
Lung-Cancer	4331.6	4730.6	5398.4	8610.0	1983.4	6707.0	**919.2**
Prostate-Tumor−1	2683.2	2695.8	2540.0	3975.8	1743.0	3330.6	**55.2**
GLI−85	1551.6	7582.2	9505.2	15939.6	5280.6	15394.6	**1287.2**
SMK-CAN−187	1389.0	5385.8	8553.2	13685.2	1500.0	10343.2	**318.8**
ALLAML	367.2	3026.4	209.8	3841.0	1225.2	613.0	**192.6**
Colon	389.8	849.6	86.0	1126.0	339.2	205.0	**31.2**
Lymphoma	1449.2	1690.4	**312.8**	2545.0	797.8	471.8	400.0
Nci9	904.0	4081.2	521.2	2110.0	734.2	678.6	**182.8**
Leukemia	1232.8	3019.8	**222.4**	3390.8	1606.4	432.8	230.0
Prostate-GE	1617.0	2522.2	**116.2**	3183.0	2234.6	674.4	218.2
Yale	81.8	412.8	129.4	198.8	69.0	169.4	**39.0**

is $O(N\times\ D)$. N and D denote the number of populations and the dimension, respectively. The time complexity required for the evaluation function used to assess the quality of the candidate solutions is $O(N)$. The time complexity for the candidate solutions to perform positional updates during the optimization search process is $O(N\times\ D)$. Therefore, the overall algorithmic complexity of the WOA is $O(N\times D\times T)$, where T denotes the maximum number of iterations.

(2) The proposed EMSWOA

In the EMSWOA algorithm proposed in this paper, the time complexity of population initialization is the same as that of the original algorithm, which is also $O(N\times D)$. The time complexity required for the evaluation of the fitness function is also $O(N)$. The time complexity for the update of candidate populations is $O(N\times D)$. The time complexity for the execution of the dynamic multiple-population strategy is $O(N\times$

Table 6
Results of mean fitness values (AVG) and standard deviation (STD) for the EMSWOA and SOTA methods.

Dataset	Measure	GOOSE	CSOFS	BCCAS	HCLDMS -PSO	MWOA	HWOAG	EMSWOA
11-Tumor	AVG	0.12827	0.11854	0.11661	0.11910	0.10125	0.12578	**0.07039**
	STD	0.00275	0.00769	0.00837	0.00761	0.01047	0.01242	0.00745
9-Tumor	AVG	0.29826	0.36357	0.32842	0.30658	0.21390	0.30697	**0.16312**
	STD	0.02584	0.01003	0.00928	0.01065	0.01568	0.05759	0.02807
Brain-Tumor−1	AVG	0.10104	0.11766	0.11069	0.11457	0.08768	0.11486	**0.03692**
	STD	0.01014	0.00288	0.00539	0.00581	0.02228	0.01448	0.00687
Brain-Tumor−2	AVG	0.14706	0.16056	0.13402	0.16587	0.12929	0.16219	**0.05602**
	STD	0.01277	0.01150	0.01401	0.00586	0.01855	0.01397	0.00902
DLBCL	AVG	0.04822	0.04982	0.03321	0.04841	0.01769	0.05881	**0.00140**
	STD	0.00555	0.00488	0.00654	0.00581	0.01257	0.00773	0.00278
Leukemia−1	AVG	0.05735	0.03414	0.04571	0.04286	0.01780	0.05166	**0.00669**
	STD	0.00704	0.01076	0.00670	0.01007	0.01170	0.00679	0.00771
Leukemia−2	AVG	0.09689	0.08703	0.08163	0.09036	0.05310	0.09092	**0.02340**
	STD	0.00854	0.00202	0.01010	0.00601	0.00891	0.00620	0.00779
Leukemia−3	AVG	0.02515	0.02386	0.01877	0.02894	0.01375	0.03258	**0.00021**
	STD	0.00611	0.00879	0.00551	0.00655	0.00916	0.00343	0.00029
Lung-Cancer	AVG	0.04812	0.04899	0.04324	0.05106	0.02978	0.04846	**0.02173**
	STD	0.00693	0.00609	0.01184	0.00351	0.00646	0.00310	0.00341
Prostate-Tumor−1	AVG	0.09457	0.09188	0.08856	0.09373	0.08506	0.09798	**0.02109**
	STD	0.00703	0.00232	0.00573	0.00585	0.01127	0.00484	0.00353
GLI−85	AVG	0.09146	0.10111	0.08908	0.10047	0.08637	0.10167	**0.03445**
	STD	0.00859	0.00590	0.01270	0.00869	0.01595	0.00735	0.01657
SMK-CAN−187	AVG	0.20506	0.29917	0.27179	0.29680	0.18569	0.30960	**0.13297**
	STD	0.00935	0.00571	0.00497	0.00797	0.01999	0.00657	0.00630
ALLAML	AVG	0.09406	0.11317	0.05636	0.12168	0.08727	0.09790	**0.00096**
	STD	0.02522	0.00070	0.02079	0.00519	0.02035	0.01287	0.00146
Colon	AVG	0.13047	0.13885	0.07500	0.14726	0.09874	0.11134	**0.02468**
	STD	0.00637	0.00703	0.00774	0.01111	0.02275	0.02225	0.00610
Lymphoma	AVG	0.04955	0.04534	0.04884	0.05256	0.04210	0.05477	**0.03999**
	STD	0.00243	0.00174	0.00528	0.00033	0.00043	0.00297	0.00387
Nci9	AVG	0.28928	0.38858	0.30272	0.32873	0.23058	0.34765	**0.16909**
	STD	0.01073	0.00681	0.02066	0.01895	0.00706	0.01925	0.01289
Leukemia	AVG	0.05638	0.06695	0.02361	0.06893	0.03308	0.03988	**0.00025**
	STD	0.01393	0.00507	0.00553	0.01034	0.00995	0.01048	0.00038
Prostate-GE	AVG	0.09745	0.08944	0.06125	0.09085	0.08594	0.08862	**0.03449**
	STD	0.00788	0.00615	0.01505	0.00333	0.00817	0.00615	0.02090
Yale	AVG	0.26381	0.34063	0.28992	0.28734	0.19731	0.30609	**0.17094**
	STD	0.00959	0.00378	0.00972	0.01563	0.00758	0.00868	0.01350

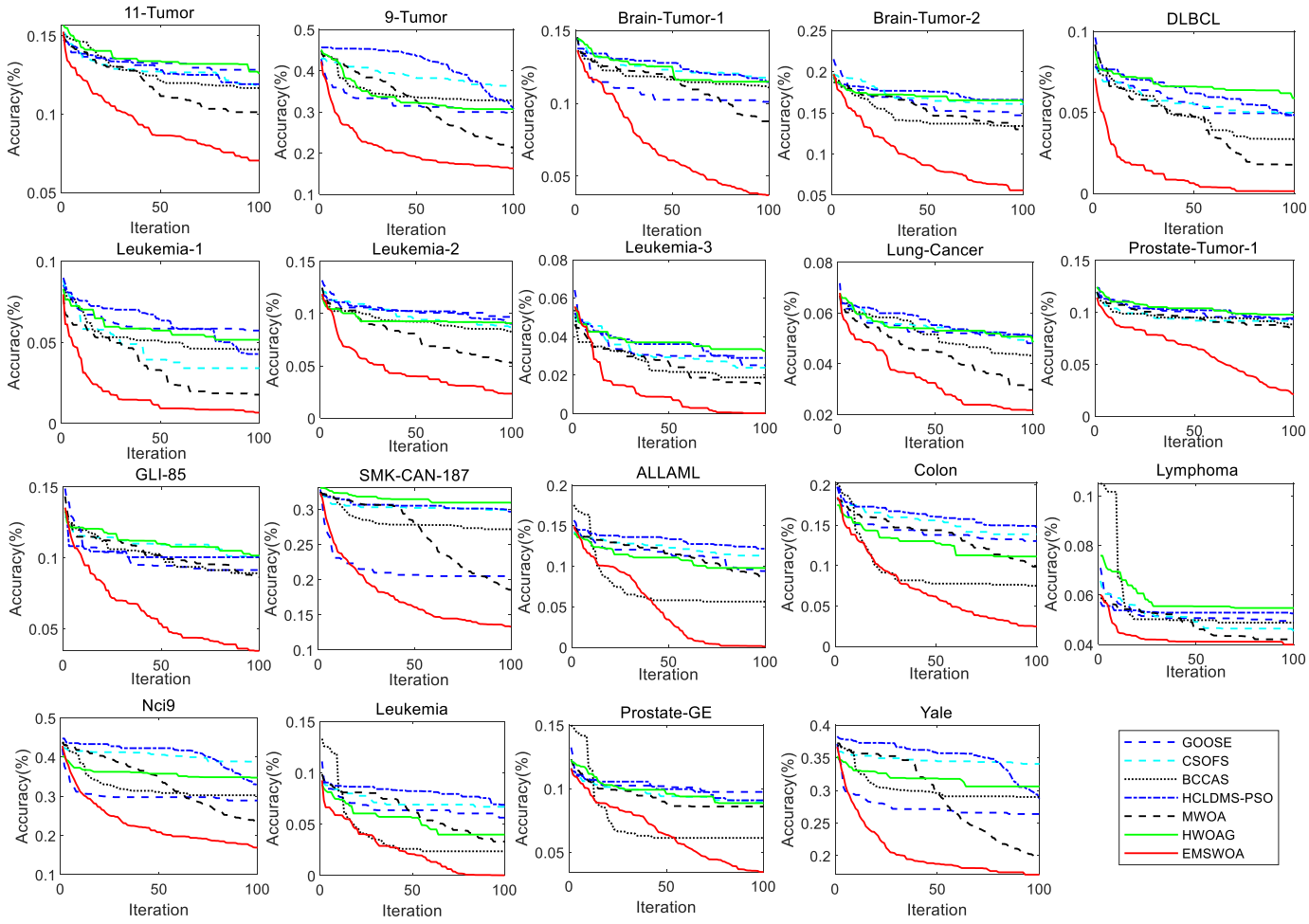


Fig. 8. Comparison of the average fitness convergence curves of the EMSWOA and SOTA methods.

D). The time complexity for the execution of the elite tuning strategy is $O(N \times D)$. Therefore, the overall time complexity of the EMSWOA at the number of iterations T is. The overall time complexity is $O(N \times D \times T)$. The algorithm possesses similar time complexity before and after improvement.

4. Experimental results and discussion

In this section, a benchmark dataset of high-dimensional FS common in the field was selected for validation. The experiments are tested and compared on two core FS metrics: accuracy and number of features. Moreover, to assess the superiority of the EMSWOA more comprehensively, the comparison algorithms chosen in this paper include both

Table 7
P value test results for the EMSWOA and state-of-the-art methods.

Dataset	GOOSE	CSOFS	BCCAS	HCLDMS-PSO	MWOA	HWOAG
11-Tumor	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
9-Tumor	0.0079	0.0079	0.0079	0.0079	0.0317	0.0079
Brain-Tumor−1	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
Brain-Tumor−2	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
DLBCL	0.0079	0.0079	0.0079	0.0079	0.0159	0.0079
Leukemia−1	0.0079	0.0079	0.0079	0.0079	**0.1508**	0.0079
Leukemia−2	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
Leukemia−3	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
Lung-Cancer	0.0079	0.0079	0.0317	0.0079	**0.0556**	0.0079
Prostate-Tumor−1	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
GLI−85	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
SMK-CAN−187	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
ALLAML	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
Colon	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
Lymphoma	0.0079	0.0079	0.0079	0.0079	0.0317	0.0079
Nci9	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
Leukemia	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079
Prostate-GE	0.0079	0.0079	**0.0556**	0.0079	0.0079	0.0079
Yale	0.0079	0.0079	0.0079	0.0079	0.0079	0.0079

evolutionary computational class algorithms and feature selection algorithms with a non-evolutionary (NON-EA) computational class.

4.1. Experimental setup

4.1.1. High-dimensional datasets

To validate the performance and generalization ability of the EMSWOA on different datasets, a total of 19 benchmark datasets [24,54] commonly used in high-dimensional feature selection performance evaluation were selected. These benchmark datasets have different categories and sample sizes, with unbalanced data between categories. Complex and diverse categories and sample distributions can more effectively evaluate the generalization capability of FS algorithms under complex distribution conditions and have the ability to identify effective feature subsets. This paper draws on a commonly used selection of datasets from extensive literature [24,54] in this field, selecting 19 benchmark datasets composed of one-dimensional feature data of type $1 \times n$, where n denotes the number of original features, as detailed in Table 2.

4.1.2. Comparison algorithms

We select two classes of comparison algorithms for analysis with the EMSWOA algorithm. The first class is the evolutionary computation class of algorithms of the same type as the WOA, which includes: two classical algorithms of the evolutionary computation class and five state-of-the-art (SOTA) approaches in the field of high-dimensional FS. The latest algorithm is the GOOSE algorithm (GOOSE) [55]. Five SOTA methods are used in the field of high-dimensional FS, including the competitive swarm optimizer feature selection algorithm (CSOFS)[56], the binary chaotic crow search algorithm (BCCAS) [57], the heterogeneous composite learning and dynamic multi-swarm particle swarm algorithm (HCLDMS-PSO) [58], the modified whale optimization algorithm (MWOA) [59], and the hybrid whale optimization algorithm (HWOAG) [16]. The relevant parameter settings are shown in Table 3.

Another class of representative FS algorithms of the non-evolutionary computation class includes discriminative FS based on structured sparse subspace learning modules (S^2DFS) [60], FS based on adaptive graph learning and constraints (EGCFS) [61], multi-criteria decision-based multi-label feature selection (MFS-MCDM) [62], and support vector machines with recursive feature elimination (SVM-RFE) [63] and its equivalent extended algorithm (MSVM-RFE) [64]and RFE-CBR [65] with correlated bias reduction and the Online group streaming feature selection based on fuzzy neighbourhood granular ball rough sets (OGSFS-FNGBRS) [66].

Table 8
Comparison of EMSWOA and WOA performance.

Dataset	WOA		EMSWOA	
	ACC±STD	Number	ACC±STD	Number
11_Tumor	90.11 ± 0.85	3427.00	**92.64 ± 1.37**	1097.00
9_Tumor	68.00 ± 3.21	242.80	**72.33 ± 4.35**	233.20
Brain_Tumor_1	91.11 ± 1.36	1066.80	**94.89 ± 1.27**	122.00
Brain_Tumor_2	87.20 ± 2.28	1854.00	**92.40 ± 1.67**	336.80
DLBCL	99.22 ± 0.71	739.00	**100.00 ± 0**	493.20
Leukemia_1	98.61 ± 0.98	989.60	**99.44 ± 0.76**	506.80
Leukemia_2	94.44 ± 1.39	1473.20	**97.50 ± 1.16**	428.20
Leukemia_3	99.44 ± 0.76	2093.80	**100.00 ± 0**	1054.40
Lung_Cancer	96.45 ± 0.22	2221.80	**98.03 ± 0.35**	919.20
Prostate_Tumor_1	92.94 ± 1.45	1007.80	**97.06 ± 0.69**	55.20
GLI_85	93.88 ± 2.10	4291.00	**96.71 ± 1.74**	1287.20
SMK-CAN−187	72.94 ± 1.76	2553.80	**75.83 ± 1.75**	318.80
ALLAML	91.11 ± 1.86	1609.80	**100.00 ± 0**	192.60
Colon	88.71 ± 1.98	698.40	**96.77 ± 1.14**	31.20
Lymphoma	95.83 ± 0.74	860.80	**96.04 ± 0.47**	400.00
Nci9	63.00 ± 2.74	427.60	**69.00 ± 2.53**	182.80
Leukemia	95.83 ± 3.26	1924.80	**100.00 ± 0**	230.00
Prostate-GE	92.16 ± 0.69	1735.60	**95.49 ± 1.32**	218.20
Yale	66.67 ± 3.95	45.20	**70.18 ± 2.89**	39.00

4.2. Experimental analysis

4.2.1. Comparison of average accuracy rates

The EMSWOA was tested with seven comparison algorithms on 19 benchmark datasets with high-dimensional feature selection. Table 4 presents the results of the experiments for average classification accuracy and standard deviation (STD), with the optimal values in bold.

First, comparing the EMSWOA algorithm to the 6 comparison algorithms overall, the EMSWOA achieved the best average classification accuracy on all 19 high-dimensional datasets. Among them, the average classification accuracy of the EMSWOA reached 100 % on the DLBCL, Leukemia_3, ALLAML and Leukemia datasets, and only the EMSWOA algorithm reached the theoretical optimum on these four datasets. Moreover, the STD of the EMSWOA algorithm is 0 on the DLBCL, Leukemia_3, ALLAML and Leukemia datasets, which indicates that the improved the EMSWOA not only has a higher accuracy in finding the best, but also the robustness of the algorithm. Secondly, compared to the best of four excellent algorithms in high-dimensional FS, the EMSWOA algorithm improves accuracy by more than 3 % on average over nine datasets, indicating that the EMSWOA is still more competitive even when compared to new high-dimensional FS approaches that have been proposed recently. Compared with the HCLDMS-PSO algorithm, which is also a multi-swarm algorithm, the EMSWOA algorithm has an average accuracy improvement of more than 6 % on eleven datasets, which indicates that the dynamic multi-population through the central distance can effectively improve the algorithm's mining ability, and is more conducive to learning for local information, which enhances the algorithm's exploitation ability. The EMSWOA algorithm obtained the highest classification accuracy on all 19 datasets, but only obtained a smaller standard deviation on 5 datasets, which may be due to the fact that the elite tuning process is affected by the current optimal solution in the iteration cycle, which increases the volatility of the EMSWOA while speeding up the convergence. Overall, the EMSWOA algorithm searches for a better subset of features and obtains higher classification accuracy.

We present a plot of the average accuracy convergence curves for the EMSWOA and the comparison algorithms on the 19 datasets, as shown in Fig. 7. Overall, the EMSWOA outperforms all the comparative algorithms in terms of convergence curves on 19 datasets, and the algorithm converges faster and with less fluctuation in the convergence curves, indicating that the EMSWOA can perform a faster search for the best solution on the high-dimensional feature selection dataset. In addition, the EMSWOA converges slightly slower than the comparison algorithms do in the early iterations on the Leukemia_3, GLI_85, SMK-CAN-187, ALLAML, Colon, Lymphoma, Nci9, Leukemia, Prostage-GE and Yale datasets, but it is superior to the comparison algorithms in terms of convergence accuracy. The reason for the above phenomenon is that the multi-swarm division in the EMSWOA algorithm initially limits the exploration space, thus restricting global search capabilities. However, the dynamic multi-swarm mechanism of the EMSWOA prevents the algorithm from being trapped in local optima because it has a single population. Therefore, with increasing iteration numbers, the EMSWOA demonstrates stronger global optimization capabilities.

4.2.2. Average number of features

The number of features selected is also an important measure of the algorithm's performance. We compared the average number of features contained in the optimal subset of features selected by the EMSWOA and the other six compared methods, and the experimental results are shown in Table 5, with the optimal values bolded.

First, the EMSWOA algorithm is compared with six comparison algorithms, and the selected feature subset is compressed by an average of a factor of 12.31. Among them, the optimal feature subset selected by the EMSWOA algorithm on 16 high-dimensional datasets has the smallest average number of features, and is only slightly lower than the BCCAS algorithm on Lymphoma, Leukemia and Prostate-GE. Second, the EMSWOA algorithm compresses features by an average of 3.58 times

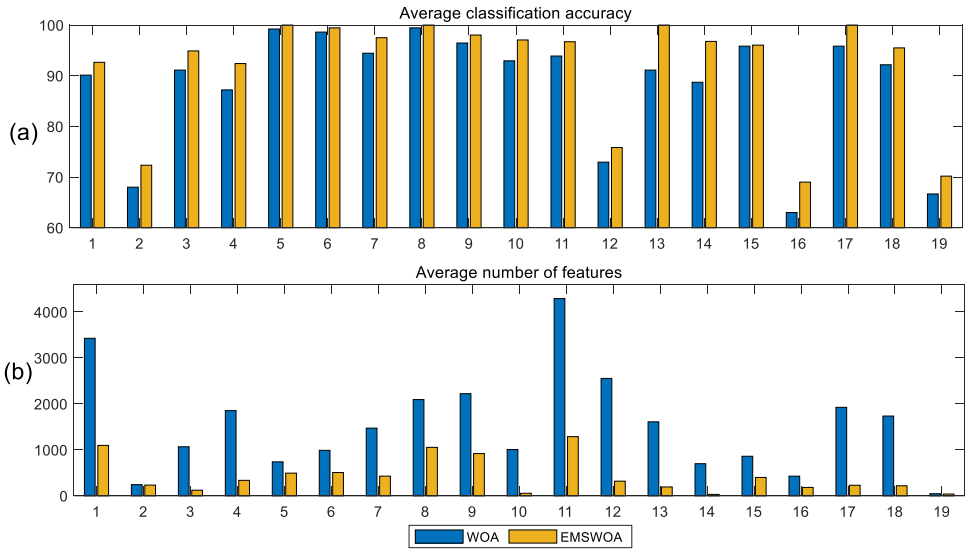


Fig. 9. Comparison of the EMSWOA and WOA performance. (a) is a histogram comparing the average classification accuracy of the EMSWOA and WOA, and (b) is a histogram comparing the number of selected features of the EMSWOA and WOA.

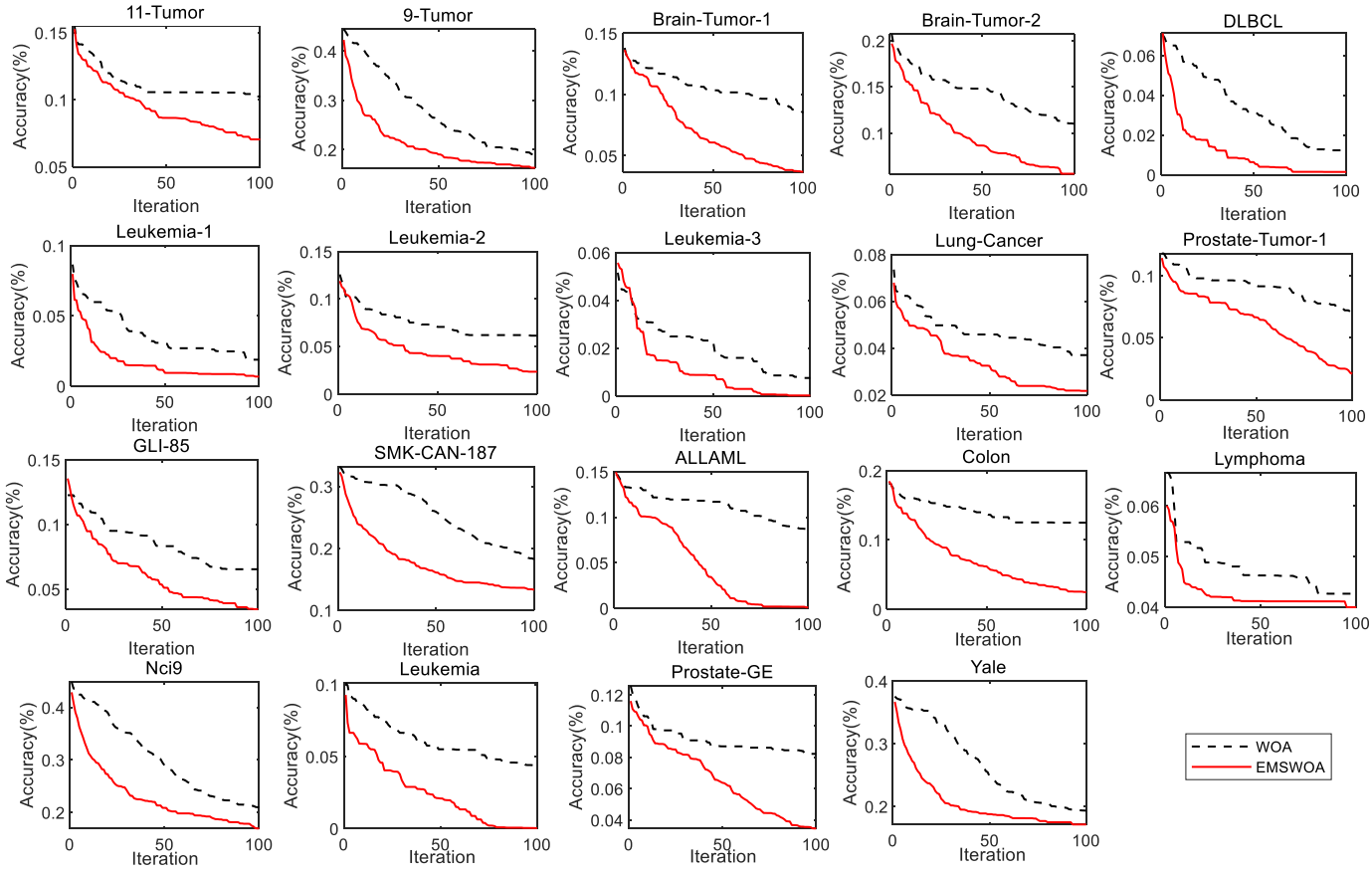


Fig. 10. Comparison of the WOA and EMSWOA fitness convergence curves.

compared to the optimum among all compared algorithms, where the EMSWOA algorithm reduces the number of optimal feature subsets by more than 13 times on 2 datasets, including 39.35 times on the Prostate-Tumor-1 dataset, suggesting that compared to the novel high-dimensional feature selection algorithms proposed in recent years, the EMSWOA is still competitive. The significant reduction in the number of features contained in the optimal feature subset of the EMSWOA algorithm is due to the fact that our proposed the elite tuning strategy effectively removes the redundant features by correcting the invalid probabilistic flip-flops using the optimal solution to obtain an optimal subset with a better number of features. As a result, the EMSWOA algorithms are more capable of identifying valid features and removing redundant features than the SOTA methods, thus increasing the accuracy of classification and reducing memory consumption.

Table 9
The average accuracy (ACC%) and the average number of selected features (NUM) of EMSWOA compared to the latest EA-based feature selection algorithms.

Dataset	Measure	MGWO -C	PPSO -EDT	OSC -EA	CE -BEGJO	MPEA -FS	BAEOAFT	EMSWOA
11_Tumor	ACC	89.75	92.02	-	-	80.98	-	**92.64**
	NUM	223.60	209.00	-	-	60.78	-	1097.00
9_Tumor	ACC	62.45	71.11	-	-	54.94	-	**72.33**
	NUM	78.40	100.30	-	-	19.61	-	233.20
Brain_Tumor1	ACC	87.19	87.89	90.20	-	84.81	-	**94.89**
	NUM	15.70	72.50	75.50	-	6.09	-	122.00
Brain_Tumor2	ACC	88.05	89.63	83.34	-	78.57	-	**92.40**
	NUM	39.80	165.20	43.00	-	5.63	-	336.80
DLBCL	ACC	98.74	98.13	98.68	-	92.36	-	**100.00**
	NUM	10.30	65.60	22.50	-	3.60	-	493.20
Leukemia_1	ACC	98.68	**99.87**	-	-	91.56	-	99.44
	NUM	25.40	41.80	-	-	3.70	-	506.8
Leukemia_2	ACC	-	94.99	-	-	89.68	-	**97.50**
	NUM	-	44.90	-	-	4.42	-	428.20
Leukemia_3	ACC	-	99.18	-	-	-	-	**100.00**
	NUM	-	146.10	-	-	-	-	1054.40
Lung_Cancer	ACC	96.63	-	-	-	-	-	**98.03**
	NUM	94.50	-	-	-	-	-	919.20
Prostate_Tumor1	ACC	-	94.49	-	-	-	-	**97.06**
	NUM	-	41.40	-	-	-	-	55.20
GLI_85	ACC	-	-	-	94.12	-	**97.26**	96.71
	NUM	-	-	-	146.15	-	12454	1287.2
SMK-CAN−187	ACC	-	-	71.49	-	-	69.10	**75.83**
	NUM	-	-	48.20	-	-	11613	318.80
ALLAML	ACC	-	-	-	99.44	-	95.58	**100.00**
	NUM	-	-	-	18.75	-	10.97	192.60
Colon	ACC	-	-	88.72	92.34	-	94.45	**96.77**
	NUM	-	-	17.30	13.40	-	1025	31.2
Lymphoma	ACC	-	-	-	-	-	88.42	**96.04**
	NUM	-	-	-	-	-	2223.5	400.0
Nci9	ACC	-	-	-	68.50	-	**76.73**	69.00
	NUM	-	-	-	92.50	-	5534.10	182.80
Leukemia	ACC	-	-	98.92	99.17	-	-	**100.00**
	NUM	-	-	30.73	16.85	-	-	230.00
Prostate-GE	ACC	-	-	-	-	-	85.00	**95.49**
	NUM	-	-	-	-	-	2906.10	218.20
Yale	ACC	-	-	-	-	-	56.67	**70.18**
	NUM	-	-	-	-	-	600.10	39.00

Table 10
Average classification accuracy (%) of the EMSWOA compared with the NON-EA feature selection algorithms.

Dataset	S^2DFS	EGCFS	MFS -MCDM	MSVM -RFE	RFE -CBR	OGSFS -FNGBRS	EMSWOA
11_Tumor	78.63(+)	85.86(+)	84.97(+)	78.64(+)	81.58(+)	-	**92.64**
9_Tumor	46.36(+)	47.94(+)	51.82(+)	52.35(+)	48.18(+)	-	**72.33**
Brain_Tumor_1	71.43(+)	82.14(+)	78.57(+)	72.73(+)	75.00(+)	-	**94.89**
Brain_Tumor_2	76.67(+)	63.33(+)	74.44(+)	79.16(+)	77.78(+)	-	**92.40**
DLBCL	91.67(+)	79.17(+)	95.83(+)	79.99(+)	83.33(+)	93.37	**100.00**
Leukemia_1	91.30(+)	73.91(+)	82.61(+)	88.85(+)	87.83(+)	-	**99.44**
Leukemia_2	83.33(+)	75.00(+)	85.23(+)	76.52(+)	74.17(+)	94.74	**97.50**
Leukemia_3	91.30(+)	95.65(+)	81.62(+)	87.08(+)	89.57(+)	-	**100.00**
Lung_Cancer	85.71(+)	90.48(+)	90.48(+)	86.17(+)	85.71(+)	90.21	**98.03**
Prostate_Tumor_1	80.65(+)	87.10(+)	77.42(+)	91.36(+)	96.77(-)	-	**97.06**
GLI_85	92.31(≈)	88.46(+)	88.46(+)	83.66(+)	84.62(+)	-	**96.71**
SMK-CAN−187	65.52(+)	65.52(+)	68.97(+)	72.80(+)	**75.86(≈)**	-	75.83

4.2.3. Fitness value analysis

The fitness value is the algorithm's comprehensive trade-off between the classification accuracy and the number of features during feature selection, which reflects the algorithm's global search ability. In this paper, we give the fitness values and standard deviations of the EMSWOA with six comparison algorithms (shown in Table 6), and the convergence graph of the corresponding fitness values, shown in Fig. 8.

As demonstrated in Table 6, the EMSWOA algorithm consistently achieved the lowest fitness values across 19 high-dimensional datasets, indicating its superior global optimization capabilities compared to six other feature selection algorithms in tasks involving complex, high-dimensional spaces. Additionally, the standard deviations (STDs) are provided, with the EMSWOA obtaining the lowest STDs on the DLBCL, Leukemia-3, Colon, and Leukemia datasets. While the algorithm's STDs on other datasets were slightly higher than the optimal values, the incorporation of multiple populations increased the diversity and perturbation of the optimal solutions found, enhancing the likelihood of achieving better outcomes. Consequently, the fitness values of EMSWOA were lower than those of the six comparison algorithms across all 19 datasets.

The convergence curves of fitness values depicted in Fig. 8 clearly show the EMSWOA's advantages over comparative algorithms. The EMSWOA not only achieved the lowest fitness values on all 19 datasets but also demonstrated faster convergence speeds compared to all

comparative algorithms. The EMSWOA performed slightly worse than the comparative algorithms during the early iterations on the 11_Tumor, Leukemia-3, GLI-85, ALLAML, Colon, Leukemia, and Prostate-GE datasets. However, it significantly outperformed the comparative algorithms as the iterations progressed, a transient effect caused by the constrained search space in the early stages due to the segmentation of multiple populations.

4.2.4. P value tests of the Wilcoxon rank-sum test

The Wilkerson P value test [67] is a non-parametric statistical method used to describe the variability between data, and we tested the EMSWOA against other comparison algorithms for P values; the results are shown in Table 7.

We set $P = 0.05$ as the threshold for the significance test used to compare the variability between the EMSWOA and other comparison algorithms. Values greater than 0.05 in Table 7 are marked in bold, and values between 0.01 and 0.05 are underlined. As shown in Table 7, EMSWOA has only three values greater than 0.05, with all the remaining values being less than 0.05. In addition, the EMSWOA has three values between 0.01 and 0.05 when compared to the MWOA, and the EMSWOA has one value between 0.01 and 0.05 when compared to the BCCAS. All remaining values were less than 0.01. Therefore, this indicates that the EMSWOA algorithm is significantly different from the comparison algorithms.

4.2.5. Performance comparison before and after algorithm improvement

To verify the effectiveness of the improvements, we conducted experiments, the results of which are shown in Table 8. Where the optimal accuracy is identified in bold with underlining, the optimal value of the average accuracy is identified in bold, and the optimal value of the average number of eigenvalues is identified with underlining.

The experimental results from Table 8 demonstrate that the EMSWOA achieved the highest average classification accuracy by identifying the optimal feature subset, underscoring the meaningfulness of the enhancements implemented in the EMSWOA overall. In terms of average accuracy, the EMSWOA outperformed the traditional WOA on 19 datasets, indicating that the enhancements from the elite tuning strategy and dynamic multi-swarm mechanism are reliable. Additionally, the EMSWOA exhibited superior performance in reducing the average number of features across these datasets, suggesting that the improvement strategies effectively facilitate feature elimination. For a clearer visual comparison of the pre- and post-improvement algorithms, histograms of their average classification accuracies and feature counts are provided in Fig. 9. Furthermore, Fig. 10 presents the convergence curves from the comparative experiments, where the EMSWOA, when compared to the WOA, demonstrates superior convergence speed and accuracy across the 19 datasets.

In summary, the experiments show that the EMSWOA benefits from its integrated elite tuning strategy and dynamic multi-swarm mechanism, both of which significantly enhance its feature selection capabilities. Specifically, the elite tuning strategy exhibits a strong capacity for effective feature elimination. Concurrently, the dynamic multi-swarm mechanism improves the quality of the best individuals during the elite correction phase by fostering greater diversity within the population, which in turn enhances the quality of elite correction. These two components work in synergy to significantly enhance the feature selection capabilities of the EMSWOA.

4.2.6. Comparison with the latest EA-based feature selection algorithms

To further validate the efficacy of the EMSWOA algorithm in high-dimensional feature selection problems, we selected seven of the most recently published algorithms that have demonstrated exceptional performance in this area. Among them, the evolutionary dual-task feature selection method based on probability-based initialization particle swarm optimization (PPSO-EDT) [25], the objective space constraint-based evolutionary algorithm (OSC-EA) [26], the copula entropy-based binary enhanced golden jackal optimization (CE- BEGJO) [27], the a decomposition-based multi-population evolutionary algorithm for high-dimensional feature selection (MPEA-FS) [30] and the binary variant of the Ali Baba and the Forty Thieves algorithm (BAFT) based on adaptive elite opposition-based AFT (BAEOAFT) [29] were published in 2024 and represent the most recent advances, while the modified gray wolf optimization with copula entropy filter methods (MGWO-C) [28] is a representative algorithm published in 2023. A comparative analysis of these seven state-of-the-art algorithms against EMSWOA is presented in Table 9. Furthermore, the average classification accuracy (ACC) and the average number of selected features (NUM) cited in Table 8 are sourced from publicly available literature, where a '-' denotes that the corresponding data were not provided in the publications.

From the comparative results in Table 9, it is evident that the feature subsets selected by the EMSWOA generally achieved higher classification accuracies. Specifically, the EMSWOA attained the highest classification accuracy in 16 out of 19 datasets, with the most significant improvement observed in the Yale dataset, where the average accuracy increased by 13.51 %. Moreover, only the EMSWOA algorithm achieved a 100 % accuracy rate in the DLBCL, Leukemia_3, ALLAML, and Leukemia datasets. The EMSWOA slightly underperformed compared to the optimal values in the Leukemia_1, GLI_85, and Nci9 datasets. Analyzing from the perspective of the average number of features, the EMSWOA achieved the highest classification accuracies with the fewest features in the Lymphoma, Prostate-GE, and Yale datasets. In other datasets, the number of features selected by the EMSWOA was higher than the minimal feature count of the comparative methods; however, excessive filtering of features can lead to the loss of discriminative attributes. The comparative methods tend to overly compress the feature set, diminishing their focus on improving accuracy and consequently losing valuable features. For instance, in the Brain_Tumor2 dataset, the optimal feature subset selected by the MPEA-FS contained an average of only 5.63 features, yet the classification accuracy was only 78.57 %, significantly lower than the 92.40 % achieved by the EMSWOA. Thanks to its elite tuning strategy, the EMSWOA effectively relearns and corrects for missing effective features, thus favoring the identification and retention of discriminative features while eliminating redundant and irrelevant ones. Therefore, compared to the latest state-of-the-art methods, the EMSWOA algorithm still exhibits strong competitiveness in high-dimensional feature selection tasks.

4.2.7. Comparison with NON-EA feature selection algorithms

To more comprehensively validate the performance of the EMSWOA on the feature selection problem, five NON-EA feature selection algorithms were also selected for comparison with the EMSWOA in this study, and the results are shown in Table 10, with the optimal values in the table bolded. The results of the compared algorithms in the table are derived from publicly published data [54].

Compared with the other six NON-EA algorithms, the EMSWOA achieves the highest classification accuracy on all 11 datasets. We compared the EMSWOA with the best of the comparison algorithms, where the average classification accuracy of the EMSWOA was improved by almost 20 % on the 9_Tumor dataset. On the Brain_Tumor1 and Brain_Tumor2 datasets, the average classification accuracy of the EMSWOA improved by at least 12.75 % or more. For the 11_Tumor, Leukemia_1, Lung_Cancer and GLI_85 datasets, the average classification accuracy of EMSWOA improved by more than 4 %. On the DLBCL and Leukemia_3 datasets, the EMSWOA achieved an average classification accuracy of 100 %, which is also an improvement of more than 4 % compared with the comparison algorithms. For Leukemia_2, the EMSWOA improved by 2.76 % compared with the optimal value in the comparison algorithm. Furthermore, the EMSWOA algorithm was the only algorithm to achieve 100 % average classification accuracy on the DLBCL and Leukemia_3 datasets compared with the comparison algorithms. In general, the EMSWOA algorithm is still a more competitive

algorithm than the NON-EA algorithm.

4.2.8. Discussion

The EMSWOA method proposed in this paper enhances the ability of the algorithm to jump out of local optimal solutions through the dynamic multi-swarm mechanism based on centroid distance, and through the mining and detection of local information by subpopulations. In addition, the quality of the candidate solutions is improved by the proposed elite tuning strategy, which corrects for the invalid probability flips during discrete space mapping, thus accelerating convergence. The experimental results also show that the EMSWOA algorithm can achieve higher classification accuracy with fewer features than the comparison methods can achieve. The EMSWOA feature selection algorithm has broader application value and can be utilized to address high-dimensional data reduction issues in the field of bioinformatics processing [3]. It can also be applied in contexts such as IoT intrusion detection [68], thereby introducing new technological options for the field of data statistics and analysis.

However, the EMSWOA also has some limitations. First, the EMSWOA is only adapted to the case of a fixed length of basic features, and is not capable of coping with complex environments where the length of features changes, and can be researched in the later stage for the problem of variable-length feature selection. In addition, the EMSWOA only performs feature selection for single-label data, and cannot be used for multi-label task, and the EMSWOA algorithm can be extended to multi-label feature selection tasks in the future.

5. Conclusions and future work

In this paper, a dynamic multi-swarm WOA based on elite tuning is proposed to solve the high-dimensional feature selection problem. In the EMSWOA, dynamic multiple populations are first constructed by designing a centroid distance metric based approach to improve the exploitation ability of the EMSWOA in the decision space while maintaining population diversity. In addition, to correct for the invalid probability flip-flop problem when converting to discrete space, we design an elite tuning strategy to improve the interference of dynamic thresholding on the feature importance measure, enhance the quality of the subset of candidate features, and improve the performance of the EMSWOA algorithm. Based on a dynamic multi-swarm mechanism and an elite tuning strategy, the experimental results of the EMSWOA on 12 real-world, high-dimensional datasets show that, compared with other superior algorithms, the EMSWOA can remove redundant and irrelevant features more effectively and improve the accuracy of classification.

The dynamic multi-swarm mechanism based on the centroid distance and elite tuning strategy in EMSWOA proposed in this paper can be used not only to improve the WOA algorithm but also to improve swarm intelligence algorithms. The newly proposed strategy enhances the optimization capabilities of algorithms, aiding in providing better search methodologies for the fields of machine learning and data science. For example, the improved swarm intelligence algorithms can be employed for optimizing neural network parameters, as well as for searching neural network architectures and processing and analyzing EEG data, among other tasks, thereby offering novel solutions for these domains.

Future work could improve the global optimization process of the WOA by combining relevant algorithmic ideas from machine learning and deep learning, such as transfer learning, to accelerate the convergence of the algorithm. In addition, the scope of application of algorithms can also be expanded through domain-specific research. For example, the scope of application of algorithms can be explored for areas such as affective computing, and a priori knowledge-guided evolutionary updating strategies can be designed for specific domains to improve feature selection performance in the target domain.

CRediT authorship contribution statement

Guanjie Yan: Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Yong Wu:** Supervision, Resources, Formal analysis. **Xiaomeng Si:** Supervision, Resources, Data curation. **Fahui Miao:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Data availability

No data was used for the research described in the article.

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